Operational Toolkit for Flood Hazard Mapping

Roadmap to empirically formulated, reach-based extrapolated, ice-jam flood hazard assessment

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Introduction and Overview

Ice-jam flooding represents one of the most complex and hazardous flood processes in cold-region river systems, particularly in large, low-slope rivers where breakup dynamics, ice accumulation, and backwater effects can produce rapid and extreme increases in water levels that far exceed those associated with open-water flooding. Unlike floods driven by rainfall or snowmelt, ice-jam floods are governed by highly stochastic interactions among river discharge, channel geometry, ice cover characteristics, and the spatial location and mechanical behavior of ice jams, which together result in water levels that are difficult to predict deterministically and that often exhibit large interannual variability for similar hydrologic conditions. Traditional flood-frequency and rating-curve methods, which rely on relatively deterministice hydraulic relationships, are therefore poorly suited to representing ice-jam flood hazard, particularly when the objective is to support flood-risk management, land-use planning, or emergency preparedness in northern communities.

The material presented here addresses this challenge through a comprehensive, physically informed, and probabilistic framework for assessing ice-jam flood hazard at both gauged and ungauged river locations. The work builds on long-standing empirical and theoretical understanding of river ice processes, while integrating observational hydrometric records, frequency analysis of extreme events, and Monte-Carlo simulation techniques to explicitly quantify uncertainty. Rather than seeking to define a single deterministic ice-jam stage for a given discharge, the approach recognizes that ice-jam flooding is better characterized by envelopes of possible outcomes and by exceedance probabilities derived from ensembles of physically plausible simulations.

Fort Simpson, located on the Mackenzie River downstream of the confluence with the Liard River, is used as a primary case study because the location exhibits characteristics that typify severe ice-jam flooding in northern rivers, including large drainage area, low channel slope, strong tributary influences, and recurrent breakup ice jams that generate substantial backwater effects. Long-term hydrometric records at Fort Simpson provide an opportunity to distinguish between open-water, intact-ice, and ice-jam hydraulic regimes, to develop frequency distributions of stage and flow specific to breakup conditions, and to calibrate probabilistic ice-jam hazard models against observed extreme events, most notably the record ice-jam flood event of 2021. These calibrated relationships then serve as the foundation for extending the analysis spatially along the river reach and for extrapolating ice-jam flood hazard to

upstream ungauged locations such as Jean Marie River, where direct observational data are limited or absent.

The overall workflow presented in this document progresses logically from conceptual foundations to applied flood mapping, beginning with an introduction to rating curves in a conceptual sense [Section 1], where the distinctions between traditional open-water rating curves and the envelope-based representations required under ice-jam conditions are established. This is followed by a description of the physical and hydrologic setting of the Mackenzie River at Fort Simpson [Section 2], including the key site characteristics and data availability needed to support subsequent analysis. Using observed gauge data from Fort Simpson [Section 3], discharge–stage relationships are then developed for both open-water and intact-ice regimes, allowing the range of possible ice-jam staging for given flows to be identified. Building on this foundation, flow and stage frequency distributions are derived using BestFit [Section 4] through extreme value analysis, providing probabilistic inputs that ensure consistency between hydrologic frequency and hydraulic response. The workflow then moves to a point-based empirical assessment of backwater staging during the 2021 ice jam event at Fort Simpson [Section 5], which serves as a physically-grounded reference case. To introduce uncertainty and probabilistic reasoning, a Monte Carlo analysis is first illustrated using Manning's equation as a simplified example [Section 6], before transitioning to a full point-based ice jam flood hazard assessment using the MOCA approach at the Fort Simpson gauge [Section 7]. This point-scale framework is subsequently extended along the river centerline to generate reach-scale ice jam flood hazard profiles for Fort Simpson [Section 8], producing longitudinal water-surface elevations associated with specified annual exceedance probabilities. Profiles can be integrated with terrain data within a geographic information system to map flood inundation extents and depths using the observed 2021 ice jam event as an example [Section 9]. Finally, the calibrated and probabilistic ice jam flood hazard framework is extrapolated upstream from the gauged Fort Simpson location to the ungauged community of Jean Marie River [Section 10], demonstrating a defensible and reproducible pathway from conceptual understanding and point observations to spatially explicit flood hazard mapping under ice-jam conditions.

1. Rating curves (conceptual)

Rating curves express the hydraulic relationship between river discharge and water level and are conventionally defined under open-water conditions, where increasing discharge corresponds to a smooth, non-linear increase in water level, forming a stable, repeatable curve when plotted in stage–discharge space, as illustrated conceptually in Figure 1. Under these conditions, hydraulic roughness is dominated

by the riverbed, flow resistance varies systematically with depth, and rating curves can be extrapolated with relatively low uncertainty.

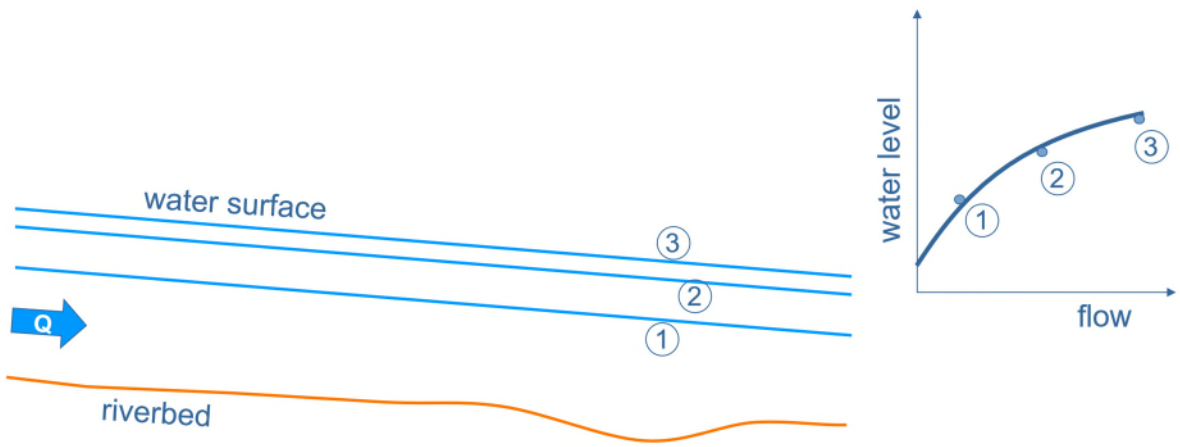


Figure 1: Conceptualisation of open-water rating curve

When an intact ice cover is present, as during mid-winter or shortly after freeze-up, the hydraulic regime changes fundamentally because a second boundary, the ice underside, contributes additional resistance to flow, resulting in greater wetted perimeter and altered velocity profiles, which causes water levels to be higher for the same discharge relative to open-water conditions, a conceptual distinction illustrated in Figure 2. Although an intact ice-cover rating curve can still be defined, it exhibits increased scatter due to spatial and temporal variability in ice roughness and thickness.

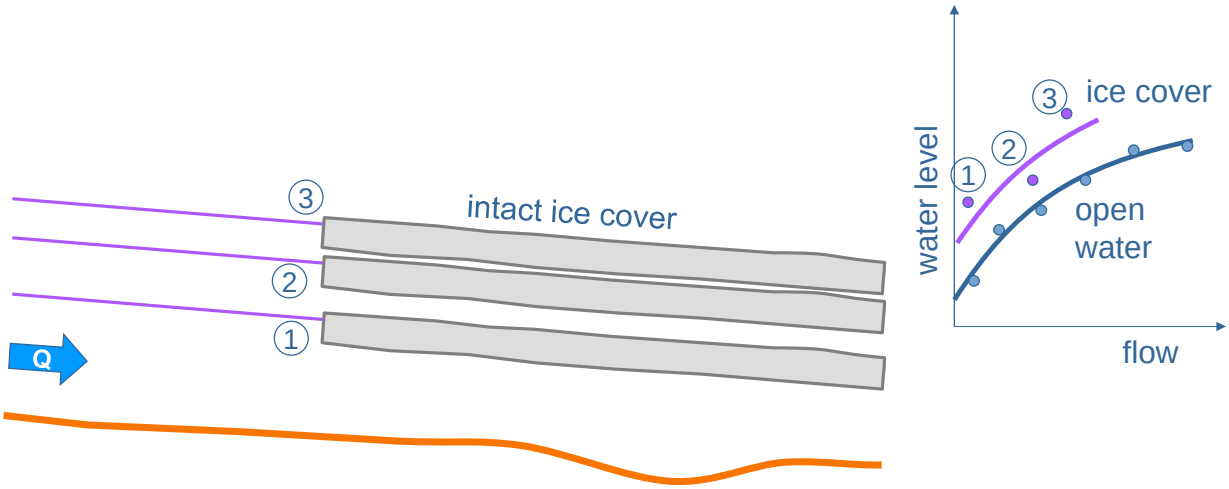


Figure 2: Conceptualisation of intact ice-cover rating curve

Ice-jam conditions depart most strongly from the traditional rating-curve concept because staged ice rubble lodges against downstream ice or morphological controls, producing highly variable backwater effects that depend not only on discharge but also on ice volume, lodgement position, ice-jam strength, and channel geometry, as shown conceptually in Figure 3. As additional ice accumulates or the ice-jam lodgement shifts downstream, water levels for the same discharge can increase or decrease abruptly, which results in a cloud of points rather than a single curve when plotted, thereby necessitating the use of envelopes rather than deterministic rating curves to describe ice-jam behavior.

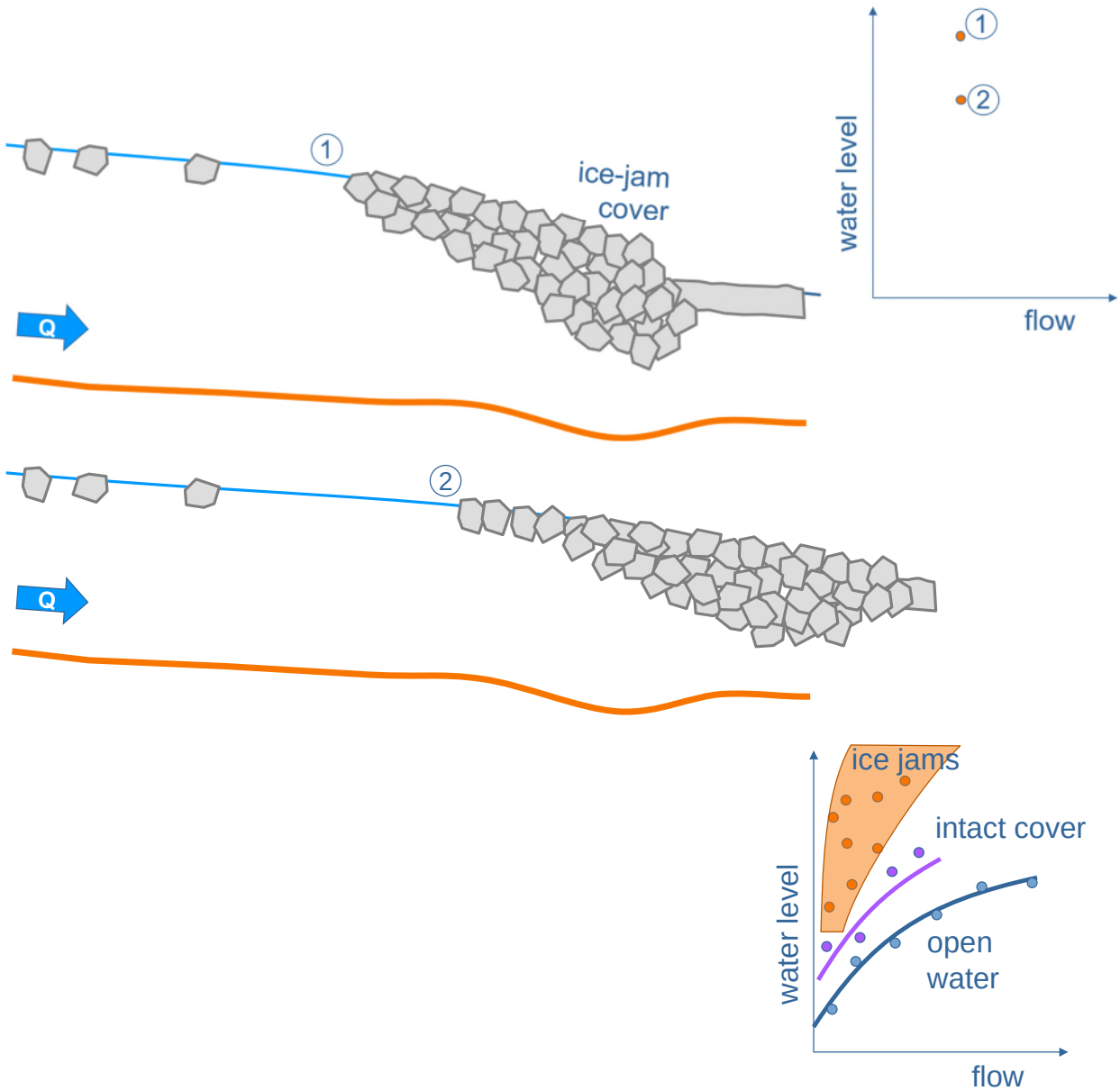


Figure 3: Conceptualisation of the ice-jam rating envelope; with all boundary conditions being equal, the backwater level of the ice jam (top panel), can rise or fall with slight shifts in ice-jam morphology, fluvial

geomorphology or ice-jam lodgement (middle panel) provides scatter in ice-jam flow/water level relationship (bottom panel).

2. Site description (Fort Simpson)

Fort Simpson is located on the Mackenzie River in southwestern Northwest Territories (Figure 4) , immediately downstream of the Liard River confluence (Figure 5) wth, a geomorphic and hydrologic configuration that exerts strong control on ice-jam formation and breakup dynamics. The Mackenzie River flows northward from Great Slave Lake, while the Liard River enters from the southwest with a steeper slope, producing asymmetrical breakup timing with the Mackenzie River breakup occurrences in most years.

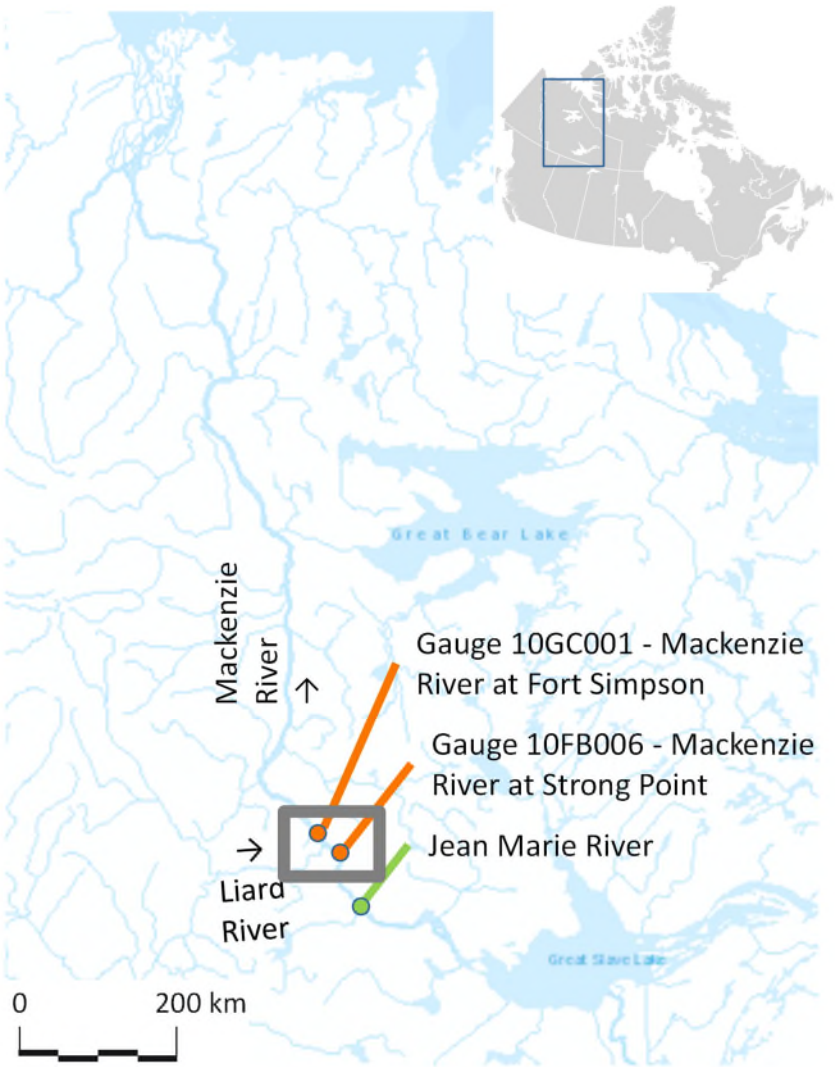


Figure 4: Location of Fort Simpson reaches at the Mackensie/Liard river confluence; gray square indicates bounds of the following figure.

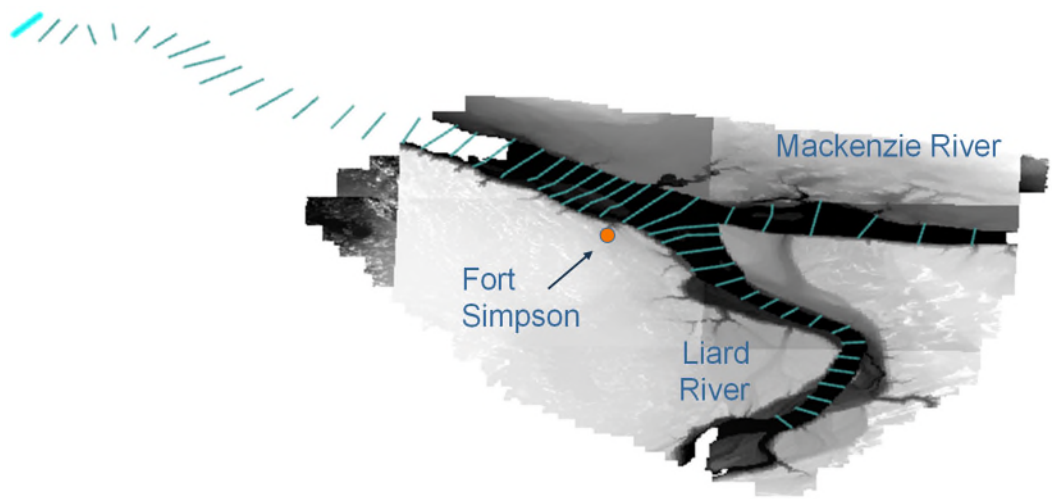


Figure 5: Location of Fort Simpson on the Mackenzie River just downstream of the rivers confluence with the Liard River.

The hydrometric gauge at Fort Simpson (10GC001) provides long-term records of discharge and water levels, while an upstream gauge at Strong Point (10FB006) captures flow contributions along the upper Mackenzie River upstream of the Mackenzie/Liard river confluence. A further upstream reach at Jean Marie River remains ungauged. Bathymetric surveys conducted by NHC established 34 cross-sections along the Mackenzie River segment and 17 along the Liard River segment (Figure 5), which later support geometric parameterisation and slope estimation.

A severe ice-jam event occurred during the spring breakup of 2021. The instantaneous water-level elevations at the gauge at Fort Simpson for the time period 1996 to mid-2023 are shown in Figure 6. The highest water-level elevation was recorded on 14 May 2021, establishing a critical benchmark for the flood-hazard analysis.

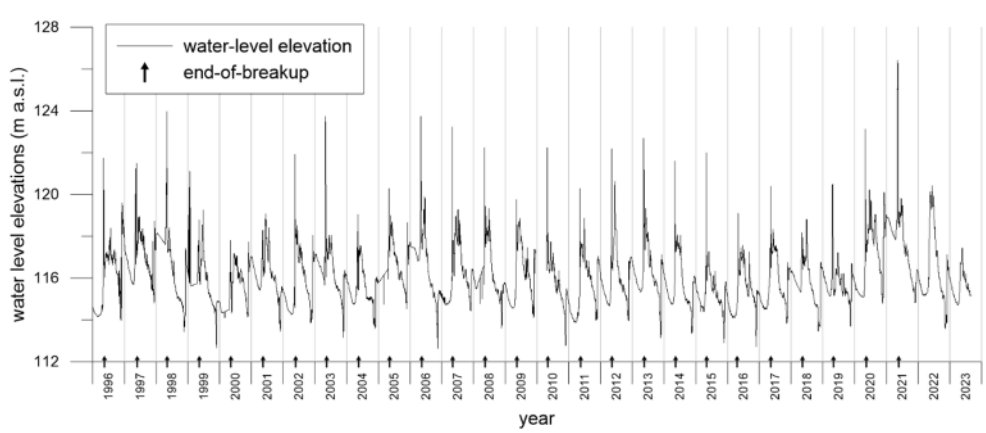
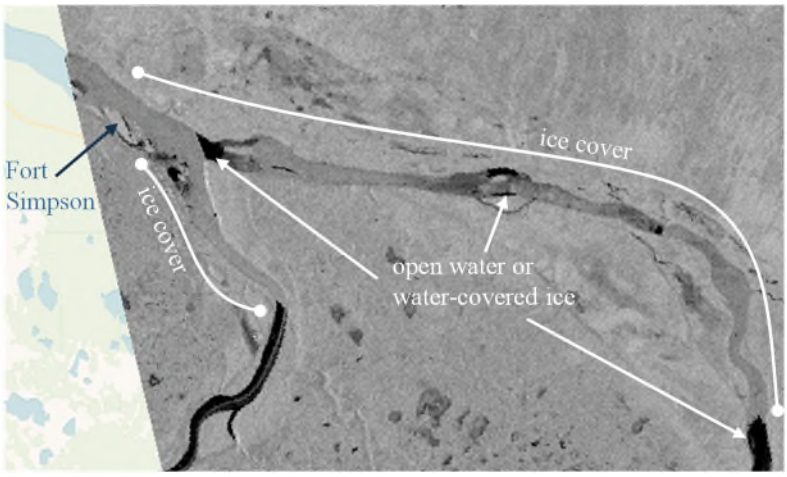


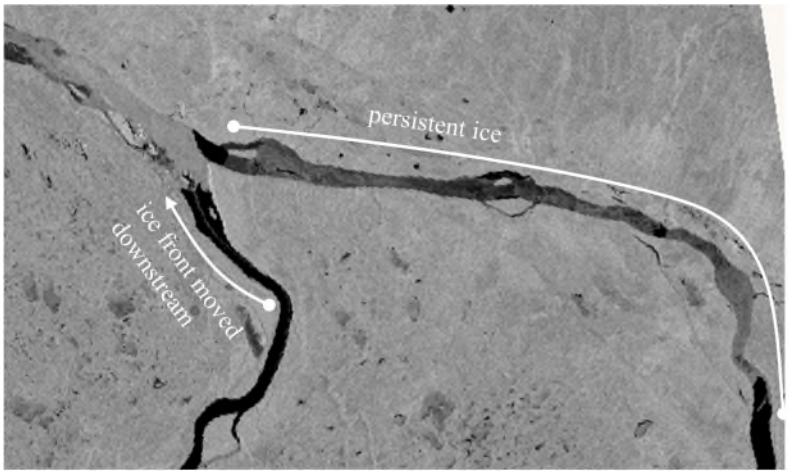
Figure 6: Hourly water level elevations recorded at Fort Simpson from 1996 to mid-2021.

Remote-sensing evidence from Sentinel-1 imagery acquired on 9, 13, and 15 May 2021 (Figure 7) shows persistent ice on the upper Mackenzie River while the Liard River had largely cleared, a recurring pattern supported by several decades of gauge records.

9 May 2021



13 May 2021



15 May 2021

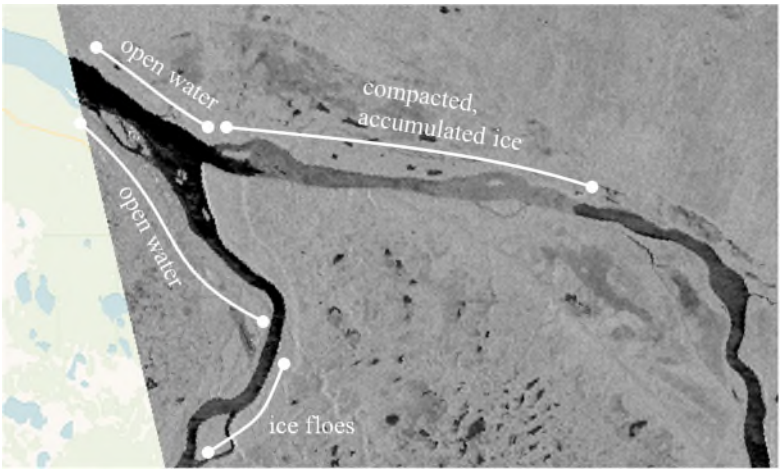


Figure 7: Sentinel-1 AWS-IW-VVVH images from 9 May 2021 (top panel), 13 May 2021 (middle panel) and 15 may 2021 (bottom panel).

3. Rating curves (Fort Simpson gauge data)

Observed discharge–stage relationships at the Fort Simpson gauge are separated into three hydraulic regimes using instantaneous and daily mean hydrometric data. An open-water rating curve is derived by pairing summer discharge values with corresponding geodetic water-surface elevations obtained by applying the CGVD2013 gauge datum offset, resulting in a smooth power-law relationship (Figure 8).

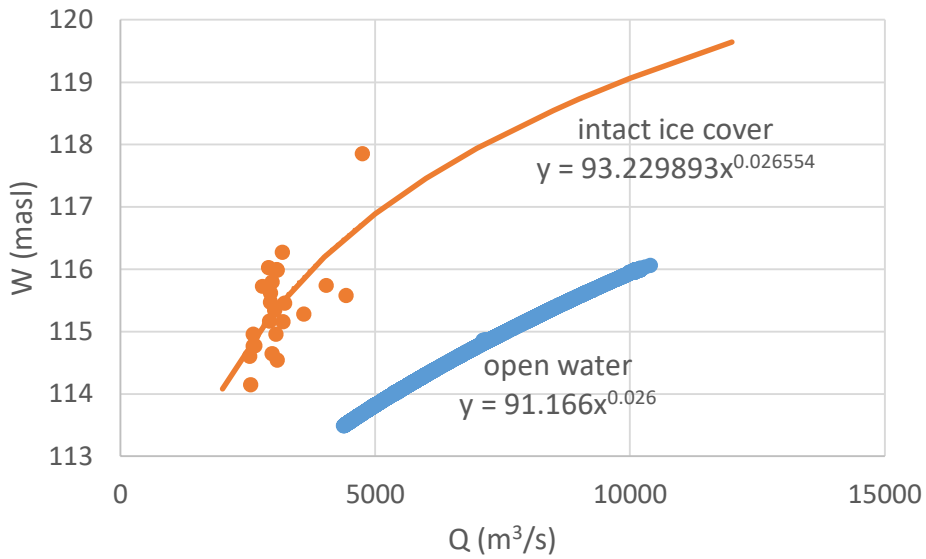


Figure 8: Open-water and intact ice-cover rating curves established at the Mackenzie River gauge at Fort Simpson.

Intact-ice rating curves were developed using under-ice discharge measurements collected during winter field surveys, in which ice thickness, flow depth, and flow velocity are measured across a cross-section at the gauge using ADCP techniques (see Figure 9), and discharges are summed across panels to which the corresponding daily mean ice-affected water levels recorded at the gauge are assigned. The resulting rating curve produces a more scattered relationship compared to the open-water rating curve (see Figure 8).

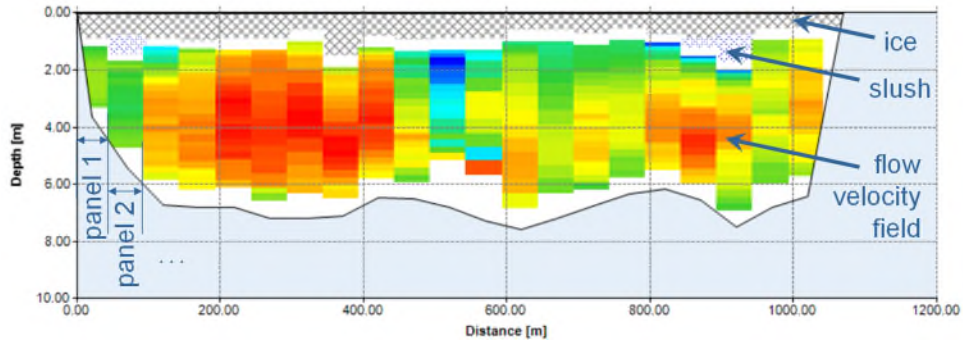


Figure 9: ADCP acquired cross-sectional profile of ice thickness, flow depths and flow velocities on 21 January 2022.

Ice-jam staging data are obtained by extracting maximum instantaneous water levels recorded during breakup for each year and associating them with daily mean discharges, recognizing that flows vary more slowly than stages during ice-jam formation. Because ice-jam points show substantial scatter and overlap at similar discharges (Figure 10), no single ice-jam rating curve is fitted, and instead an envelope with upper and lower bounds is constructed to bound observed staging, reflecting the inherently stochastic nature of ice-jam hydraulics. It also indicates the different regimes of ice flooding during breakup, including fully-developed ice jams, partially-developed ice jams and thermal breakup (see Figure 11), partitioned by NHC (2025) to 59%, 27.5% and 13.5%, respectively.

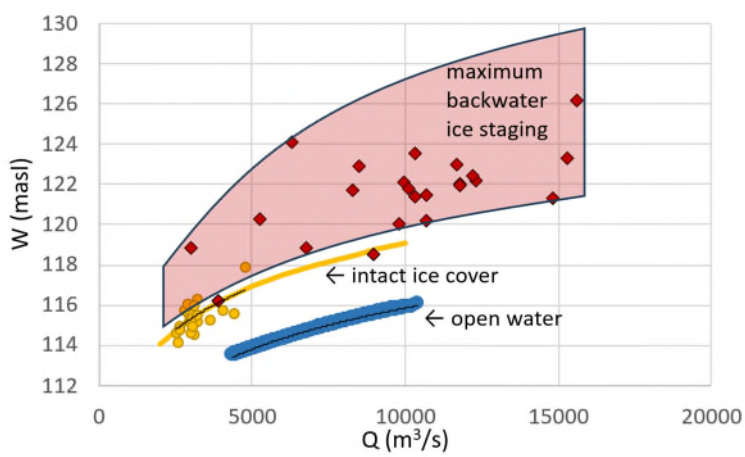


Figure 10: An envelope of peak backwater staging during ice-cover breakup alongside the intact ice-cover and open-water rating curves.

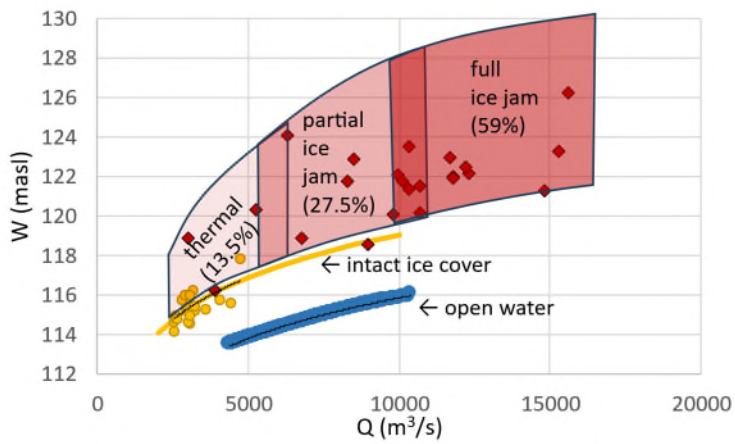


Figure 11: The envelope partitioned to different breakup types: fully developed ice jam, partially developed ice jam and thermal breakup (partitions and overlap provided for conceptual context only and may not be to scale).

4. Flow and stage frequency distributions (using BestFit)

Flow and stage frequency distributions for ice-jam events at Fort Simpson were developed using the RMC BestFit software (https://www.rmc.usace.army.mil/Software/RMC-BestFit/). Annual peak ice-jam discharges and corresponding peak stages are imported as paired datasets after removing years affected by gauge failures or incomplete records.

Multiple probability distributions were evaluated using objective goodness-of-fit criteria, and the Generalized Extreme Value (GEV) distribution is selected because of its consistent performance across evaluation metrics and suitability for modelling extremes. The function can be expressed as (Turkann, 2014a, 2014b):

$$F(x) = 1 - e^{-\left[1+\xi 1\cdot\left(\frac{x-\mu 1}{\sigma 1}\right)\right]^{-\frac{1}{\xi 1}}} \qquad 4.1$$

where $\mu 1$, $\sigma 1$ and $\xi 1$ are the GEV parameters of location, scale and shape, respectively. The fitted GEV parameters are transferred to Microsoft® Excel® and later applied to generate synthetic ensembles of flows and stages via the inverse GEV function, where random probabilities are drawn from a uniform distribution. The inverse formulation *z(p)* of Equation 4.1 becomes (Turkann, 2014a):

$$z(p) = \mu 1 - \frac{\sigma 1}{\xi 1}\cdot\left[1 - (-\ln(p))^{-\xi 1}\right] \qquad 4.2$$

where p is a random number from a uniform distribution between 0 and 1. In Excel, values for $0 < p < 1$ can be generated using the RAND() function.

The frequency distribution of the daily mean flows during peak ice-induced staging is shown in Figure 12. These synthetic flow ensembles form the probabilistic discharge input for the Monte-Carlo simulations.

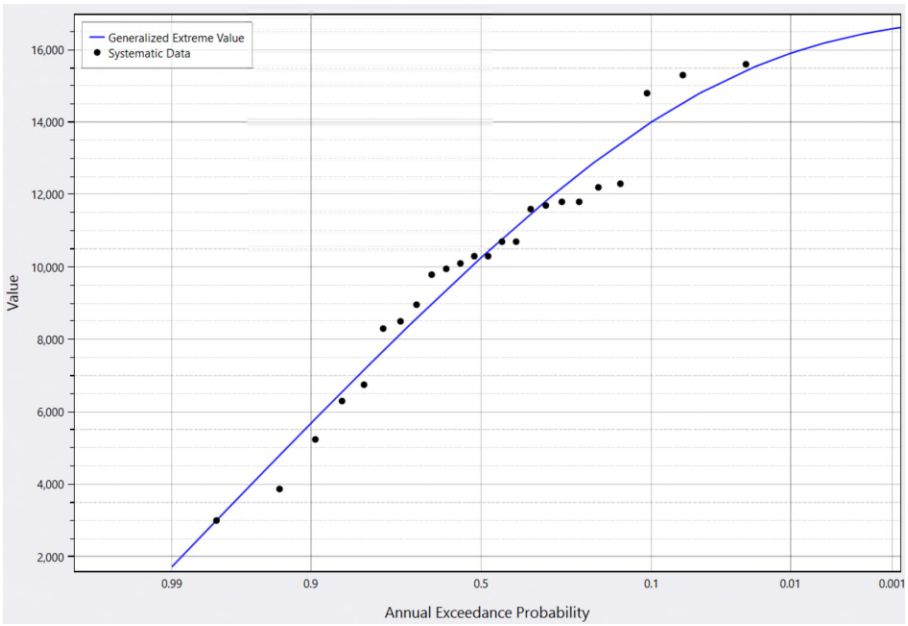


Figure 12: Frequency distribution of daily mean flows occurring during peak backwater staging at spring breakup at the Mackenzie River gauge at Fort Simpson.

Frequency distributions of the instantaneous water levels corresponding to the flows are provided in Figure 13. The three best fitting distributions, GEV, log-Pearson type 3 and Pearson type 3 provide a span of 1:100 and 1:200 AEP levels used for the calibration of the Monte-Carlo Analysis framework described below.

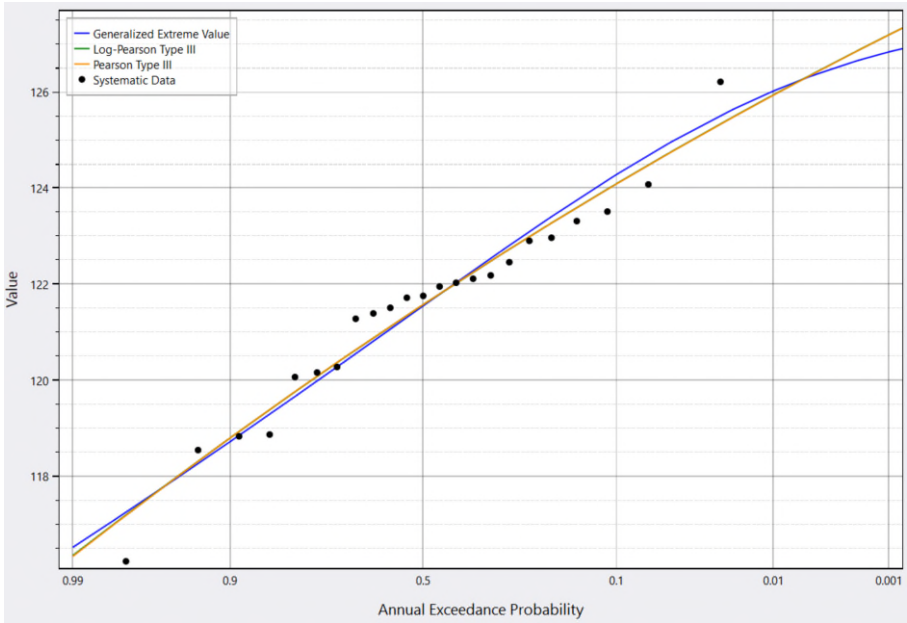


Figure 13: Frequency distribution of ice-induced peak backwater staging at spring breakup at the Mackenzie River gauge at Fort Simpson.

5. (Point) backwater staging of the 2021 ice-jam event (empirical approach)

In this section, backwater staging caused by an ice jam is demonstrated using an empirical approach that is intentionally chosen for its simplicity when compared to a full numerical river-ice model but nonetheless grounded in river-ice hydraulics. Rather than solving conservation equations through spatially discretized modelling, this method relies on empirical relationships that capture the dominant physical controls on ice-jam backwater. The empirical formulations recognize that ice-jam backwater arises from the interaction between flowing water and an accumulated mass of ice that partially obstructs the channel, and that the resulting water-level increase upstream of the jam is governed by a combination of hydraulic, geometric, morphological, and ice-mechanical factors (Beltaos, 1983, 1995).

Referring to Figure 14, when an ice jam forms, the magnitude of backwater staging H is influenced by hydraulic variables such as the river discharge Q, which determines the volume and momentum of water flowing towards the ice jam, as well as channel width W and channel slope S, exert strong control on both the formation and persistence of ice jams and on the spatial extent of backwater effects. Jam-related variables are equally important and include the depth of flow beneath the ice jam h, which controls hydraulic resistance, and the thickness of the jam t, which affects both resistance and load transfer within the ice rubble.

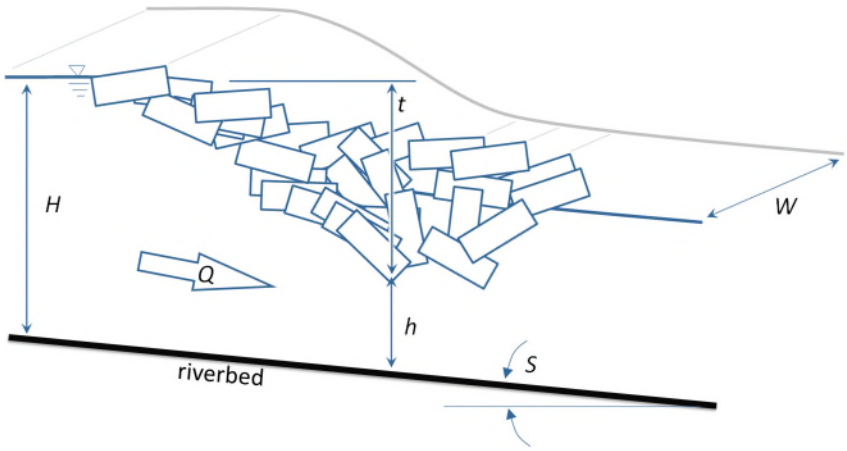


Figure 14: Conceptualisation of an ice jam.

The empirical approach itself is structured around two central relationships, one describing a non-dimensional measure of discharge ξ and the other relating this discharge to a measure of non-dimensional stage η:

$$\xi = \frac{\left(\frac{q^2}{gS}\right)^{1/3}}{WS} \tag{5.1}$$

$$\eta = \frac{H}{WS} \tag{5.2}$$

The non-dimensional discharge ξ is defined using the unit discharge q, obtained by dividing the total discharge Q by the channel width W ($q = Q/W$) and then normalizing this value by the product of gravitational acceleration g and channel slope S. This expression is raised to the power of one-third and further normalized by the product of width and slope, which means that channel width and slope appear multiple times in the denominator of the formulation. As a result, even relatively small uncertainties in width or slope can translate into substantial differences in calculated backwater, making these parameters particularly influential and emphasizing the importance of their careful estimation.

The non-dimensional stage η is likewise expressed as a function of channel width and slope and can be related to the non-dimensional discharge ξ through a relationship plotted on log–log axes (see Figure 15). Similarly to the scatter observed in open-water rating curves, intact-ice rating curves, and ice-jam stage envelopes, this non-dimensional relationship does not collapse to a single line but instead spans a range of plausible outcomes, reflecting natural variability in ice processes and channel conditions. This spread

reflectsthe features of ice-jam hydraulics rather than a deficiency of the method, and it provides a conceptual basis for later probabilistic treatment.

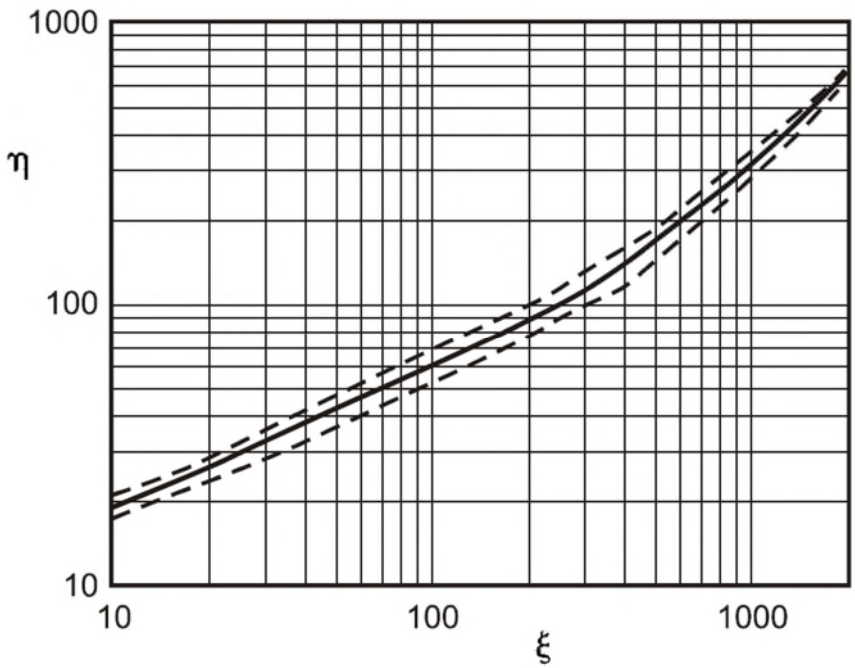


Figure 15: Relationship between non-dimensional discharge ξ and non-dimensional ice-jam stage η (Beltaos, 1983)

The non-dimensional stage η can be extended to include three empirical parameters that represent ice-related resistance and mechanical behavior:

$$\eta = \frac{H}{WS} = 0.63 f_o^{1/3} \xi + \frac{5.75}{\mu}\left(1 + \sqrt{1 + 0.11 \mu f_o^{1/3}\left(\frac{f_i}{f_o}\right)\xi}\right) \qquad 5.3$$

The first of these is the composite friction factor f_0 which represents an effective average of the friction factor associated with the riverbed and that associated with the underside of the ice cover. As illustrated in Figure 16, shear stresses develop simultaneously along the riverbed and along the ice-cover underside with each contributing to the overall flow resistance. The presence of these two rough boundaries fundamentally alters the vertical velocity profile beneath the ice, such that maximum velocities tend to occur closer to the centre of the flow rather than near the water surface, as is the case for open-water conditions.

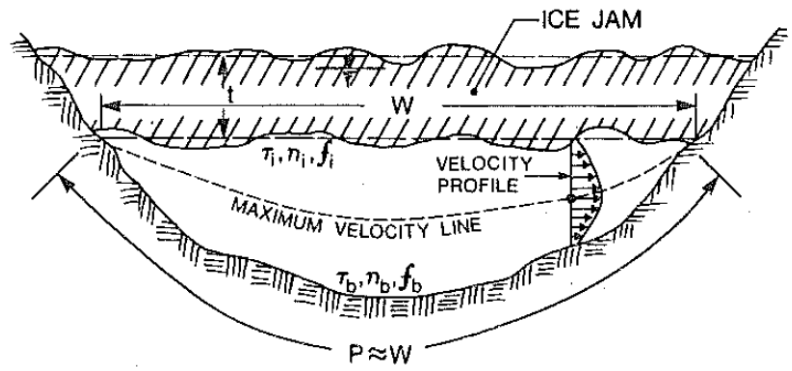


Figure 16: Shear stress τ, Manning's roughness n and friction factor f for the riverbed (subscript b) and ice-cover underside (subscript i); from Beltaos (1983).

The friction factor for ice f_i is formulated using the Darcy-Weisbach equation and is directly related to shear stress τ_i, fluid density ρ_i and mean velocity V_i:

$$f_i = \frac{8\tau_i}{\rho V_i^2} \qquad 5.4$$

Typical values of the ice-related friction factor are commonly reported to fall in the range of approximately 0.10 to 0.80, values which are generally larger than Manning's roughness coefficients for ice n_i which are often half or a quarter less, ranging approximately 0.05 to 0.20, depending on the specific range of conditions considered.

The second empirical parameter in the formulation is the ratio of the ice friction factor to the composite friction factor f_i/f_o, which controls how resistance is partitioned between the ice and overall flow boundaries. The third parameter, denoted as μ, represents the internal strength of the ice jam and accounts for a suite of mechanical processes including lateral stresses within the ice jam, friction between the ice cover and riverbanks, cohesion and freezing within the rubble mass, and the overall thickness of the ice jam. Physically, this parameter governs how an ice jam responds to incoming ice, with weaker ice jams tending to shove and thicken more readily, while stronger ice jams resist deformation and instead veer additional ice accumulation farther upstream.

Empirical ranges exist for all three parameters, although their degree of constraint varies (see Table 1). The composite friction factor f_o can often be guided by channel slope, with steeper rivers generally exhibiting higher effective friction due to increased turbulence and roughness interactions, while

low-slope rivers tend toward lower values. Mildly-slope rivers, such as large lowland systems, may therefore exhibit composite friction factors near the lower end of the reported range. The remaining two parameters typically show weaker or less direct correlations with easily measured quantities and are therefore more difficult to constrain *a priori*. Among the three, the composite friction factor is usually the most sensitive parameter in the backwater calculation, making it a natural focal point for calibration.

Table 1: Typical ranges of f_o, f_i/f_o and μ for different riverbed slopes (Beltaos, 1983)

$0.09 < f_o < 0.67$

$0.63 < f_i/f_o < 1.62$

$0.6 < \mu < 1.3\ (3.5)$

$0.11 < f_i < 0.73$

Location (1)	Q, in cubic meters per second (4)	W, in meters (5)	S, in meters per kilometer (6)	H, in meters (7)	t, in meters (8)	f_o (9)	f_i/f_o (10)	μ (11)	ξ (12)	μ (13)	f_i
Smoky R. near mouth	210	227	0.72	5.5	3.1	0.65	1.12	1.0	30.5	33.8	0.73
Smoky R. near mouth	283	228	0.72	6.2	3.3	0.55	1.17	1.0	36.9	37.9	0.64
Smoky R. near Hunting Creek	456	163	0.83	5.9	1.2	0.46	0.63	3.5	73.0	43.6	0.29
Wapiti R. near Grande Prairie	185	153	1.02	5.8	3.1	0.67	0.97	1.0	33.8	37.2	0.65
Thames R. near Thamesville	108	41	0.77	3.9	0.9	0.13	0.81	1.0	308	125	0.11
Thames R. above Sherman Brown Bridge	196	74	0.26	4.7	0.9	0.09	1.44	1.2	728	244	0.13
Thames R. at mouth	195	106	0.13	4.6	0.7	0.09	1.44	1.3	1,005	334	0.13
Athabasca R. near Ft. McMurray	930	560	0.36	8.2	4.4	0.38	1.43	1.0	45.7	40.7	0.54
	1,100	560	0.36	7.4	3.2	0.33	1.42	1.3	51.1	36.7	0.47
	800	410	0.35	8.1	4.0	0.31	1.64	0.9	72.2	56.4	0.51
	600	405	0.35	7.4	4.0	0.32	1.62	0.8	60.8	52.2	0.52
	1,150	680	0.30	8.1	3.8	0.40	1.30	1.2	48.6	39.7	0.52
Athabaska R. near	450	193	0.75	8.7	4.7	0.56	1.27	0.6	62.5	60.1	0.71

$0.05 < n_i < 0.2$

$0.11 < f_i < 0.73$

To apply this empirical method to the 2021 ice-jam event, as shown in the video "(Point) backwater staging of the 2021 ice jam event (empirical approach)", an Excel-based worksheet is used in which the governing equations are implemented explicitly, allowing each step of the calculation to be followed transparently. Discharge and geometric inputs are taken from a calibrated RIVICE model where available (see Figure 17), of the ice-jam event of 2021 which is characterised by a discharge of approximately 15,600 m^3/s. Channel width and flow area were extracted directly from the modelling output, while channel slope was derived from the bathymetric survey data.

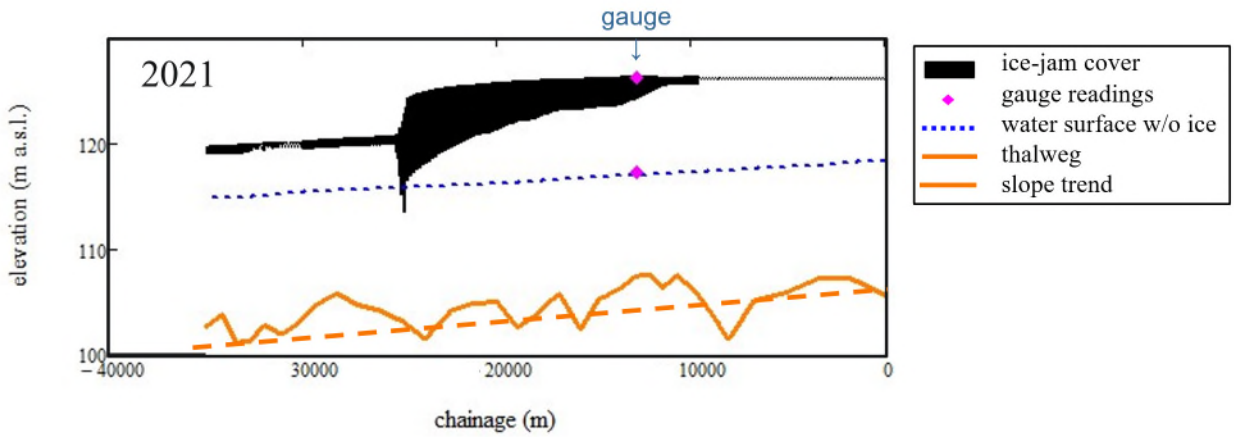


Figure 17: Longitudinal profile of the 2021 ice jam along the Fort Simpson reach of the Mackenzie River.

In cases where numerical model output or detailed bathymetry are not available, the required geometric parameters can be estimated using alternative sources. Channel width can be measured from aerial photographs or satellite imagery using tools such as Google Maps, while channel slope can be approximated from digital elevation models by comparing water-surface elevations over sufficiently long reaches. Although current, established LiDAR applications do not penetrate water, continuous water-surface elevations along a river corridor can still be extracted from terrain data, allowing slope to be estimated from elevation differences over distance. In the present case, analysis over a ten-kilometre reach indicated a water-surface elevation drop of approximately 0.7 metres, corresponding to a slope of 0.00007 (see Figure 18), while bathymetric data suggested a slightly steeper bed slope of approximately 0.000113, which was adopted for the calculations here.

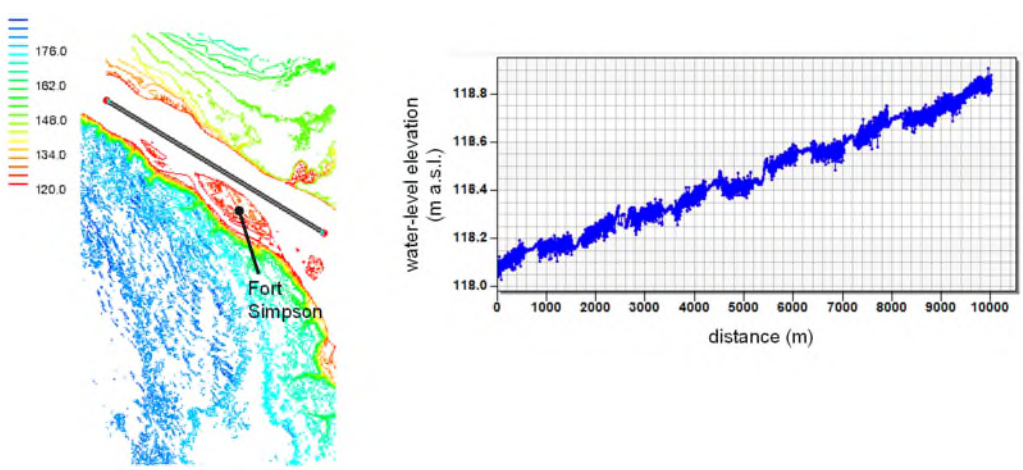


Figure 18: Determining slope along the Mackenzie River from the digital terrain model at Fort Simpson.

With these inputs defined, the non-dimensional discharge is computed using the discharge, width, and slope, and the non-dimensional stage is then calculated using the empirical expression involving the composite friction factor, the friction-factor ratio, and the ice-jam strength parameter. From the non-dimensional stage, the backwater depth H is obtained. The primary quantity of interest for hazard assessment, however, is the backwater elevation W_{peak}, which is determined by adding the computed backwater depth to the thalweg elevation.

For the 2021 event, by adjusting the empirical parameters, most notably the composite friction factor, the calculated backwater elevation W_{peak} can be brought into closer agreement with the observations, clearly demonstrating the sensitivity of the method to the total frictional representation of the river cross-section. While calibration based on a single event is limiting, this exercise illustrates the overall procedure and highlights how parameter adjustment affects results. The same procedure can be repeated for other events (not carried out here but left to the reader to experiment with), such as the 2019 and 2020 ice jams (see Figure 19), and applied across multiple years to establish plausible ranges for the empirical parameters. These ranges become particularly valuable when the method is embedded within a Monte-Carlo framework, where variability in ice-jam characteristics and associated uncertainty must be represented probabilistically rather than by single deterministic values.

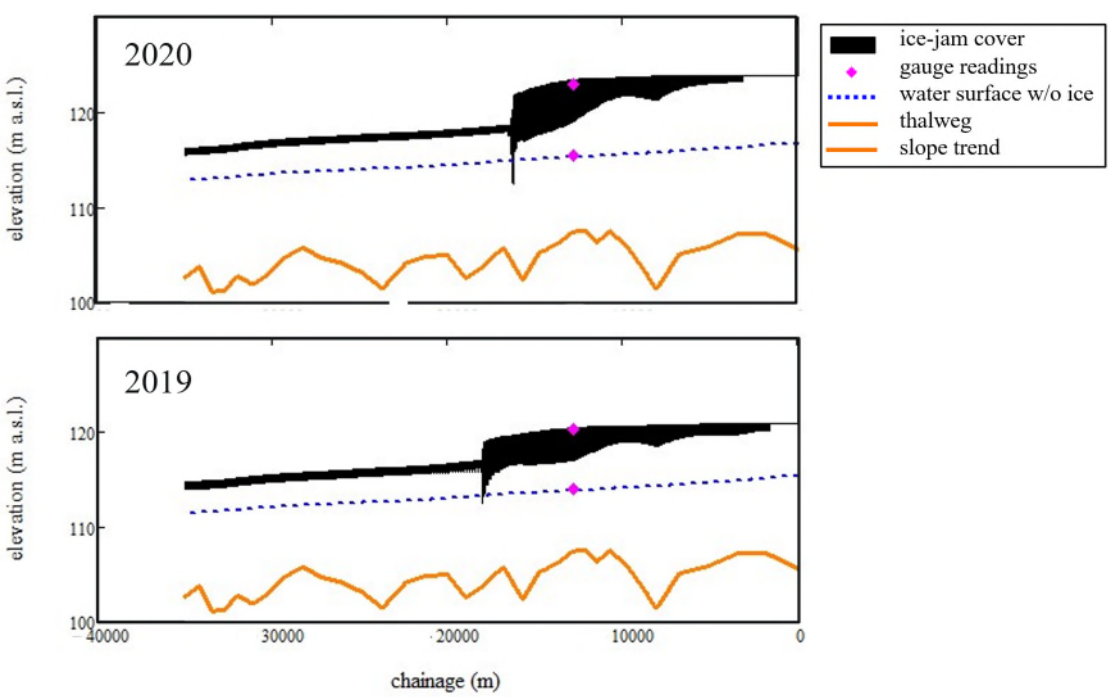


Figure 19: Longitudinal profiles of the 2019 (bottom panel) and 2020 (top panel) ice-jam events along the Fort Simpson reach of the Mackenzie River.

6. Monte-Carlo Analysis (Manning's equation example)

The next step from the previous section's backwater calculations of single ice-jam events is to carry out a Monte-Carlo analysis in which the same empirical formulations are calculated many thousands of times, with each calculation's parameter set drawn randomly from frequency distributions. This yields an ensemble and frequency distribution of backwater levels from which flood threshold elevations, particularly the 1:100 and 1:200 AEP levels, can be extracted.

To illustrate this concept in a more transparent manner, a Monte Carlo analysis is first demonstrated here for a simpler case using only Manning's equation, where discharge Q is expressed as a function of flow area A, hydraulic radius R_h, slope S, and composite roughness n_o:

$$Q = A\frac{1}{n_o}{R_h}^{2/3}\sqrt{S} \qquad 6.1$$

Under ice covered conditions, composite roughness is calculated using the Sabaneev relationship, which combines riverbed and ice roughness, respectively n_b and n_i, into a single effective coefficient:

$$n_o = \left(\frac{{n_b}^{3/2} + {n_i}^{3/2}}{2}\right)^{2/3} \qquad 6.2$$

In Monte Carlo mode, the model is run many hundreds of times, for example on the order of 500 simulations, with each simulation representing a distinct but physically plausible realization of ice-jam conditions. During each run, model parameters are randomly sampled from predefined probability distributions rather than held fixed at single deterministic values, so that, for instance, a friction-related parameter or velocity coefficient may be randomly selected within a specified range. This random sampling is applied consistently across all uncertain parameters, each of which is represented by an appropriate probability distribution reflecting either observed variability or uncertainty.

Certain variables, particularly discharge, are commonly described by extreme-value distributions such as the Gumbel or the Generalized Extreme Value distribution, reflecting their role in representing rare events, whereas other parameters, including roughness coefficients or ice-cover properties, are typically represented using uniform distributions when detailed information is unavailable or when the objective is to span a plausible range of values without introducing unnecessary bias. For each Monte Carlo realisation, the model draws one random value from each input distribution and computes the resulting

water-surface profile, and after several hundred simulations, an ensemble of profiles is produced that collectively describes the range of possible ice-jam backwater conditions.

From this ensemble, statistical percentiles can be extracted to define water-surface profiles corresponding to specific annual exceedance probabilities, such that the 1:100 AEP ice-jam flood is represented by the 99th percentile profile, meaning that 99% of simulated profiles fall below that elevation and 1% exceed it, while the 1:200 AEP event corresponds to the 99.5^{th} percentile. These percentile-based profiles can then be mapped onto the floodplain, allowing flood-hazard assessments to focus not only on observed historical events, such as the 2021 ice-jam flood, but also on rarer, higher-consequence scenarios that may not yet have occurred within the period of record.

The Monte Carlo analysis is demonstrated using the "ManningsEquation_MOCA.xlsx" Excel file and can be followed in the "Monte-Carlo Analysis (Manning's equation example)" video. The spreadsheet is organized with columns representing cross-sectional area A, composite Manning's roughness, hydraulic radius, slope, and calculated discharge, and representative values derived from the 2021 ice-jam event used to establish a baseline calculation (see Figure 20). The hydraulic radius R_h is defined as the ratio of flow area A to wetted perimeter P_w, hence $R_h = A/P_w$, which increases under ice-covered conditions because the underside of the ice contributes additional wetted boundary, and as a consequence the hydraulic radius under ice is typically smaller than for open-water conditions with the same flow area. Using approximate values of A = 20,000 m^2, a composite roughness n_o = 0.044, a hydraulic radius $R_h \approx$ 6 m and a slope S = 0.000113, Manning's equation yields a discharge of roughly 15,900 m^3/s, which is close to the observed conditions during the 2021 event and confirms that the assumed parameters are physically reasonable.

Referring to the conceptualisation in Figure 21, to move from a single deterministic calculation to a Monte Carlo framework, ranges are then assigned to selected input parameters, such that area may vary between approximately 19,000 and 26,000 m^2, (taking a window between chainages 10,000 and 15,000 m) hydraulic radius between 5 and 7 m, while slope is held constant. Random samples are generated using Excel's random number function RANDBETWEEN(), with appropriate scaling applied to achieve the desired decimal precision. These random values are propagated through the equations to produce a large ensemble of discharge calculations. Histograms are subsequently generated to visualize both the distributions of the input variables and the resulting output flows, and as the number of simulations increases, the output distributions become smoother and more stable.

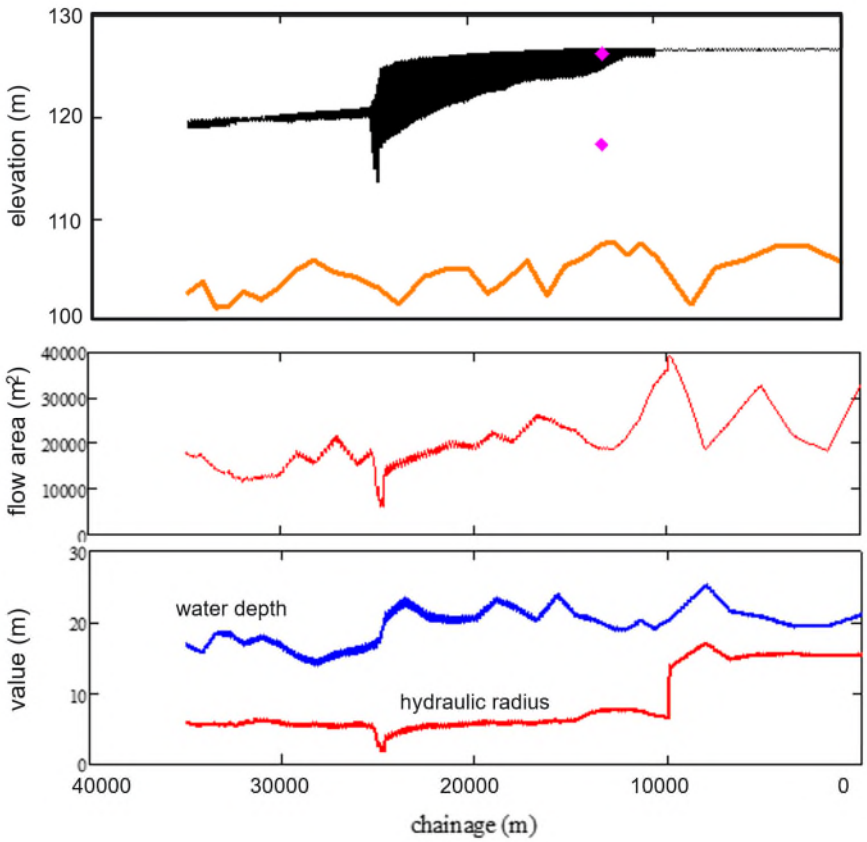


Figure 20: Profiles of backwater levels and thalweg (top panel), flow area (middle panel) and water depths and hydraulic radii (bottom panel) for the 2021 ice jam.

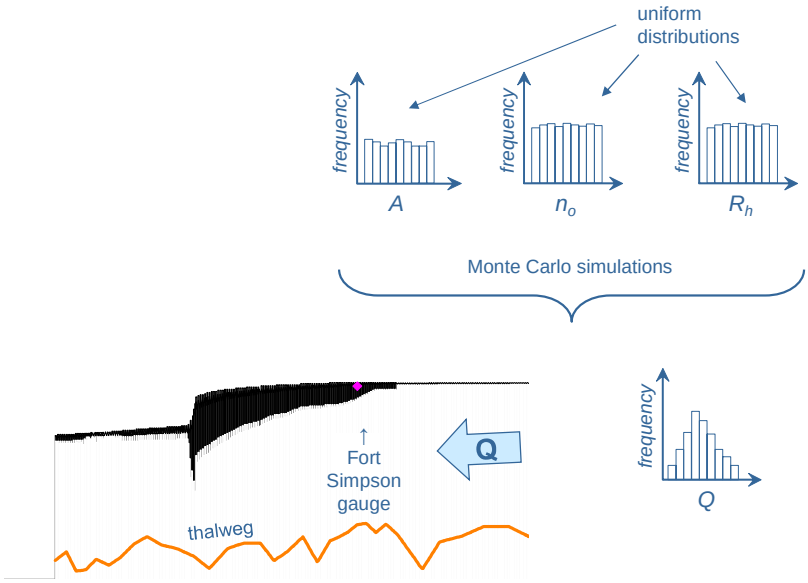


Figure 21: Conceptualisation of Monte-Carlo framework using Manning's equation and its variables (uniform distribution) as a basis.

Referring now to the conceptualisation in Figure 22, this demonstration is extended to show how uniform input distributions can be replaced with normal distributions by specifying means and standard deviations rather than minimum and maximum values, using Excel's inverse normal function NORM.INV() in combination with a random probability generator RAND(), which results in bell-shaped distributions concentrated around expected values instead of evenly spanning the range. These input distributions can be calibrated by inspecting histograms and adjusting means and standard deviations so that the majority of values fall within physically reasonable bounds, such as ensuring that hydraulic radius values remain consistent with observed geometry.

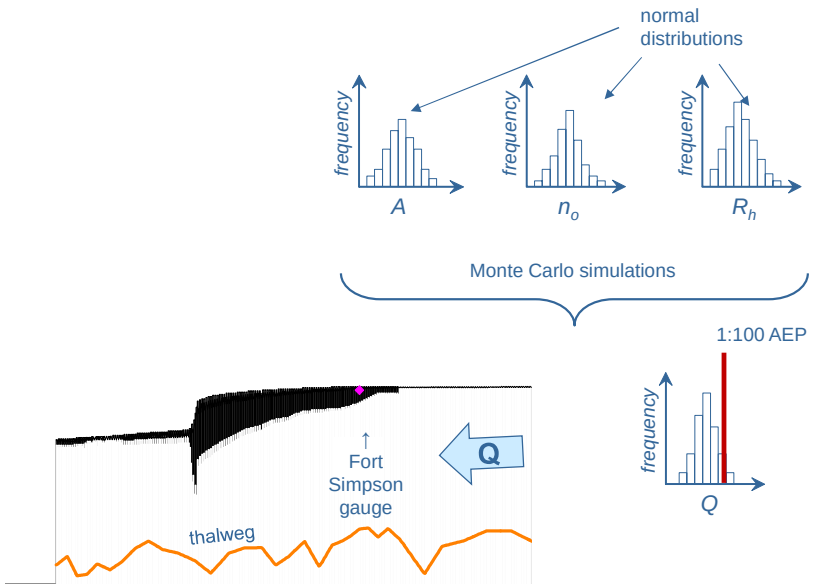


Figure 22: Conceptualisation of Monte-Carlo framework using Manning's equation and its normally-distributed variables yielding a distribution of flows from which the 1:100 AEP can be read.

The resulting simulated flow distribution is finally compared to observed or estimated return-period flows, with the 1:100 AEP event corresponding to the 99th percentile of the simulated discharge ensemble, and if this percentile is too high or too low relative to estimates obtained from the gauge data, input parameters such as composite friction factor can be adjusted accordingly. This calibration process therefore targets entire distributions rather than individual values, ensuring that the probabilistic behaviour of the model aligns with observed frequency information.

Once calibrated, the Monte Carlo framework provides a quantitative description of uncertainty in water-surface elevation that is directly applicable to flood-hazard assessment, where the goal is not to predict a single outcome but to characterize a range of plausible outcomes and their associated probabilities. It is emphasized that input variables are not required to follow normal distributions, as some, particularly extreme flows, are more appropriately described by skewed extreme-value

distributions such as the Gumbel or GEV distribution. The choice of distribution should always reflect both the physical nature of the variable and the available data.

7. (Point) ice-jam flood hazard (MOCA approach)

Here the ice-jam flood hazard at a specific point, specifically at the Fort Simpson gauge along the Mackenzie River, is analyzed using the MOCA approach. This analysis builds on previous sections in which a point backwater-stage analysis was conducted for the 2021 ice-jam event in Section 5 above, and a separate example demonstrating the Monte Carlo analysis using a simplified formulation was carried out in Section 6. In this section, the Monte Carlo framework is applied directly within Excel by using the backwater staging calculations developed for the 2021 event to estimate ice-jam flood hazard at a point.

The analysis begins by reopening the staging calibration Excel spreadsheet used previously for the 2021 event *(1) ice-jam_staging_calibration (FortSimpson).xlsx*. (the completed exercise from this section can be found in *(2) ice-jam_staging_calibration (FortSimpson).xlsx*). Following the video *"(7) (Point) ice-jam flood hazard (MOCA approach).mp4*", the focus is on the Fort Simpson gauge location along the Mackenzie River, approximately at chainage 12,800, located along the Mackenzie River channel (Figure 5). Rather than limiting the analysis to a single fixed location, the flood hazard is assessed over a broader extent of the reach stemming from this point, extending approximately from chainage 10,000 to 15,000 (see Figure 17 for orientation). This range reflects the realistic spatial variability of potential ice-jam formation zones. In the original event-based analysis, the chainage value was fixed at 12,800 m. In contrast, the present analysis transitions to a probabilistic framework by introducing randomisation, meaning the analysis is no longer tied to a single observed event. Specifically, the chainage value is treated as a random variable. This is implemented by modifying the "Ranges" worksheet so that the minimum and maximum chainage values are set to 10,000 and 15,000, respectively (see Cells B2 and B3 in Figure 23).

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	cross-section #	chainage (m)	width (m)			fo				μ (mu)	fi/fo		fi
2	1	10000	1550			0.09				0.6	0.8		0.072
3	34	15000	1600			0.11				0.8	1		0.11
4													
5													

ranges | values | geomorph | HWmarks | +

Figure 23: Sheet "Ranges" in Excel sheet.

Random chainage values are generated using Excel's RANDBETWEEN() function, with the minimum and maximum values referenced from fixed cells using absolute references. This ensures that the correct bounds are preserved when formulas are copied down the spreadsheet (see Column B of the "values" worksheet depicted in Figure 24). Channel width (Column C in Figure 24) is then determined automatically as a function of chainage by referencing the "geomorph" worksheet (see Figure 25), which contains tabulated channel geometry (e.g. width in Column B in Figure 25). A graph of the width profile along the entire reach in indicated in the left panel of Figure 26. As a result, each randomized chainage value is associated with a corresponding channel width through a LOOKUP() operation. Channel slope, derived from the thalweg elevations (see right panel of Figure 26) is held constant and referenced directly from the value defined earlier in the worksheet.

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	chainage (m)	width (m)	slope (-)	Q (m^3/s)	fo	q (m^2/s)	h (m)	**ξ (xi)**	μ (mu)	fi/fo	**η (eta)**	H (m)	t (m)		thalweg elevation (masl)	W empirical (masl)		T	p		%ile	calibrated		observed (max)	observed (min)
2	12542	1199.2	0.000113	16648.3	0.094	13.883	15.99	411.92	0.674	0.861	157.23	21.31	5.78		105.109	126.415		500	0.002	0.2%	0.998	**126.790**		126.761	126.427
3	10754	1686.5	0.000113	16397.4	0.090	9.723	12.43	230.99	0.719	0.902	96.369	18.37	6.45		105.311	123.676		200	0.005	0.5%	0.995	**126.416**		126.36	126.086
4	12623	1178.5	0.000113	16328.6	0.106	13.855	16.62	418.60	0.642	0.825	165.48	22.04	5.89		105.100	127.137		100	0.010	1.0%	0.990	**126.026**		126.017	125.791
5	14830	1497.3	0.000113	16279.9	0.110	10.873	14.32	280.31	0.692	0.989	120.62	20.41	6.62		104.850	125.258		50	0.020	2.0%	0.980	**125.324**		125.631	125.455
6	10020	1689.6	0.000113	16220.6	0.096	9.600	12.59	228.62	0.601	0.886	101.07	19.3	7.29		105.394	124.690		20	0.050	5.0%	0.950	**124.585**		125.184	125.062
7	10413	1750.5	0.000113	16147.5	0.102	9.225	12.51	214.87	0.675	0.947	95.951	18.98	7.03		105.349	124.329		10	0.100	10.0%	0.900	**123.845**		125.024	124.92
8	11166	1676.1	0.000113	16109.2	0.104	9.611	12.94	230.64	0.639	0.805	101.37	19.2	6.80		105.265	124.464		5	0.200	20.0%	0.800	**122.978**		124.456	124.409

ranges | values | geomorph | HWmarks

Figure 24: Sheet "values" in Excel file.

	A	B	C	D	E	F	G
1	chainage (m)	width (m)	open-water profile	slope (-)	slope_calc	thalweg	thalweg (trend)
2	0	2094.4	114.944	0.000050	0.00013241	105.59	106.526
3	10	2089.1	114.944	0.000050	0.00013237	105.599	106.525
4	20	2083.7	114.943	0.000050	0.00013232	105.608	106.524
5	30	2078.3	114.942	0.000075	0.00013227	105.618	106.522
6	40	2072.9	114.942	0.000075	0.00013222	105.626	106.521
7	50	2067.5	114.941	0.000050	0.00013217	105.635	106.520
8	60	2061.7	114.94	0.000075	0.00013212	105.645	106.519
9	70	2055.5	114.94	0.000075	0.00013207	105.654	106.518
10	80	2049.2	114.939	0.000050	0.00013202	105.662	106.517

ranges | values | geomorph | HWmarks

Figure 25: Sheet "geomorph" in Excel file.

Discharge values are introduced from a separate spreadsheet (Sheet "Q_GEV(random)" in the Excel file *rating curves prep.xlsx*) that contains a probabilistic flow distribution. This distribution was previously developed using Best-Fit frequency analysis of daily mean flows associated with peak ice-jam staging, as demonstrated in Section 4. A set of 1,000 random flow values is selected from the random flow generation worksheet in either Columns D or I shown in Figure 27. The selected range is intentionally broad so that negative values are extremely unlikely. However, if negative values do occur, they can be

removed through taking absolute values (Column D) or addressed by sorting and filtering (Column I). In practice, any residual error introduced by this approach is negligible and does not materially affect the uncertainty estimates. The random flow values are consistently referenced from a single column and linked back to the main calculation sheet “values”. Absolute references are removed so that, when formulas are filled down or across, the full set of 1,000 Monte Carlo realisations is generated correctly.

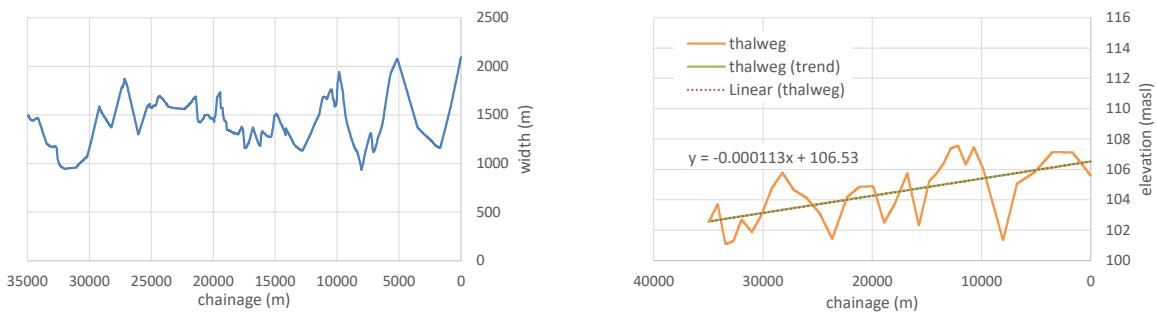


Figure 26: Profiles of width (left panel) and thalweg elevation (right panel) found in the “geomorph” worksheet.

	A	B	C	D	E	F	G	H	I	J
1	μ1	σ1	ξ1	invCDF		F		rank	inCDF	Q_sort
2	9104.50	3410.38	-0.4334	7575.712		0.7783		782	7575.712	16558.85
3				7628.147		0.7739		781	7628.147	16489.6
4				10238.948		0.5025		514	10238.95	16389.54
5				15509.480		0.0204		23	15509.48	16277.31
6				15090.014		0.0362		33	15090.01	16263.71
7				8737.879		0.6707		676	8737.879	16166.27
8				10656.114		0.4525		467	10656.11	16135.8
9				14504.108		0.0666		59	14504.11	16100.96
10				11005.557		0.4104		422	11005.56	16091.41
11				15585.604		0.0181		20	15585.6	16077.76

open water | intact ice | ice jam | Q_GEV | Q_GEV(random)

Figure 27: Sheet “Q_GEV(random) in Excel file *rating curves prep.xlsx*.

Three key model parameters are randomized in the “values” worksheet (see Figure 24): composite friction factor f_o (Column F), ratio of ice friction to composite friction f_i /f_o (Column K), and internal ice strength parameter μ (Column J) between minimum/maximum ranges set in the “ranges” worksheet (see Figure 23). Random values for each parameter are generated using the same approach as for chainage but to increase numerical precision to three decimal places (RANDBETWEEN() works only for integer values), the minimum and maximum values are first multiplied by 1000, and then the resulting value is divided by 1000. Referring to the “values” worksheet (Figure 24), unit discharge q (Column G) is calculated as discharge (Column E) divided by channel width (Column C). Depth under the ice jam h

parameter and the nondimensional discharge ξ are computed in Columns H and I, respectively. From these quantities, nondimensional stage η is calculated (Column L), which is then used to estimate backwater depth H and ice-cover thickness t, respectively in Columns M and N, for each realisation.

The thalweg elevation in Column P is obtained through a LOOKUP() call from the "geomorph" worksheet based on chainage. Because the geometry data are tabulated at 10-unit intervals (see Column A in the "geomorph" worksheet), the chainage values are implicitly rounded to the nearest interval. While this discretization could be refined further, it is sufficient for the purposes of this demonstration. Combining thalweg elevation (Column P) with backwater depth (Column M) yields the final water surface elevation W (Column Q) for each Monte Carlo realisation.

Rather than calibrating the model to a single observed water level, the results are evaluated against the return periods calculated from observed peak water-level elevations during ice jamming, previously developed from the rating-curve analysis in Section 4. Minimum and maximum bounds of the best three distribution fits are imported, and the calculated hazard curve is compared against these envelopes. Once all 1,000 realizations are generated by filling down the spreadsheet, the resulting hazard curve stabilizes and exhibits reasonable behavior. The resulting graph is shown in Figure 28.

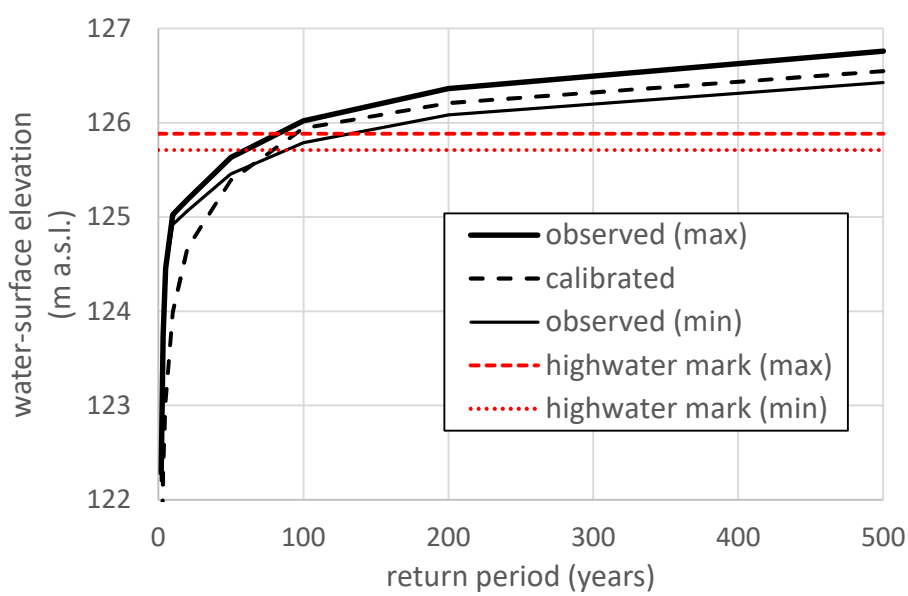


Figure 28: Ice-jam flood hazard curve.

Further refinement is achieved by adjusting the parameter ranges in the "ranges" worksheet, particularly by fine-tuning the minimum and maximum values of f_o. Additional adjustments of f_i /f_o and μ improve

agreement with the observed hazard envelopes. Some variability remains at lower return periods due to random sampling, which is expected within a Monte Carlo framework.

With calibration complete, the resulting parameterisation is deemed suitable for extrapolation to ungauged sites. In a subsequent analyse (Section 10), such an extrapolation will be applied to the Jean Marie River reach to estimate ice-jam flood hazard in the absence of bathymetric and gauge data.

8. (Reach) ice-jam flood hazard (Fort Simpson)

This section explains how ice-jam flood hazard is computed for an extended river reach at Fort Simpson rather than at a single point, building directly on the previous section's results. In addition to extending the spatial scope of the analysis, a graphical user interface (GUI) is introduced that enables the execution of many thousands of Monte Carlo simulations, making it possible to rapidly evaluate ice-jam backwater effects along an entire river reach.

Before introducing the Python-based graphical user interface (GUI), the concept will be demonstrated using the ice-jam staging Excel file *(2) ice-jam_staging_calibration (FortSimpson).xlsx* or an alternative version, *(2) ice-jam_staging_calibration (FortSimpson)_numbered.xlsx*, in which the column headers in worksheet "values" are explicitly numbered with capitalised Roman numerals, to correspond to the numbering in the workflow conceptualisation provided in Figure 29. Random values of discharge are again drawn from the GEV frequency distribution of flows during peak ice-jam staging (see [I] in Figure 29; also the actually GEV distribution in Figure 30). The slope remains a constant at 0.000113 (see [II] in Figure 29). One marked different to the point analysis is that chainage is randomized over the entire reach domain, not just between chainages 10,1000 and 15,000 but now from chainage 1 to 34960 m (see [III] in Figure 29). Widths are now looked up, using the LOOKUP() function, in worksheet "geomorph", to the corresponding randomly chosen chainage values.

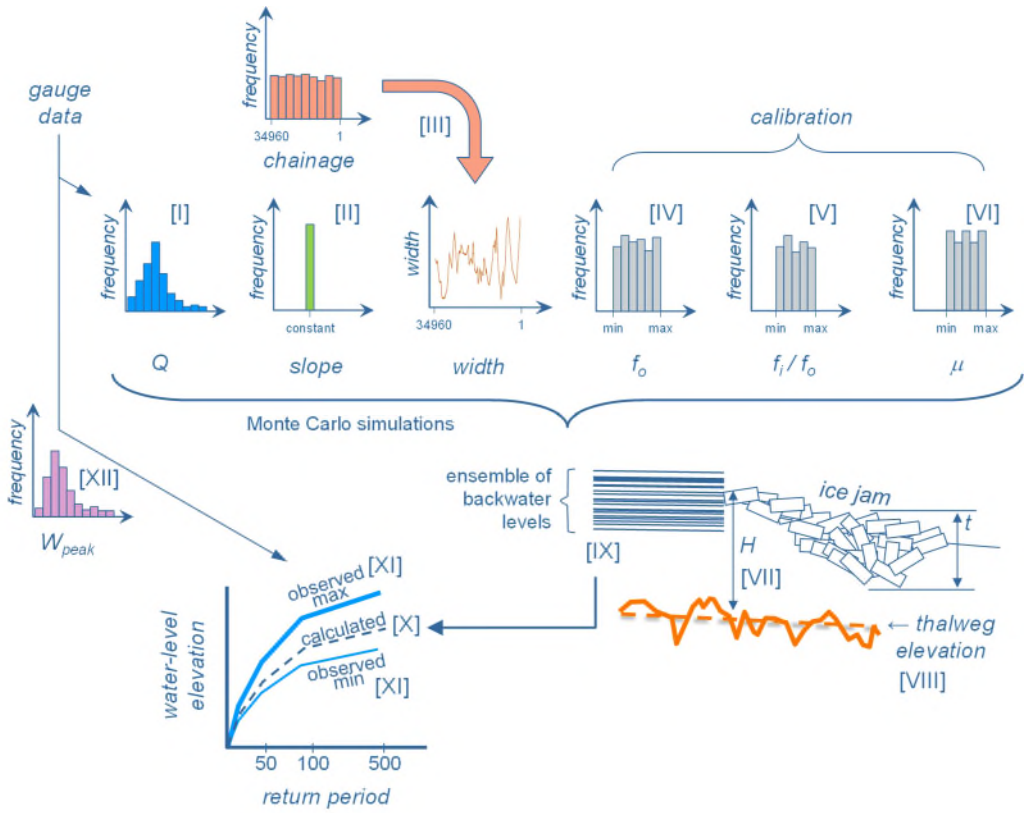


Figure 29: Conceptualisation for carrying out the ice-jam flood hazard assessment for the entire Fort Simpson reach, not just at the gauge location.

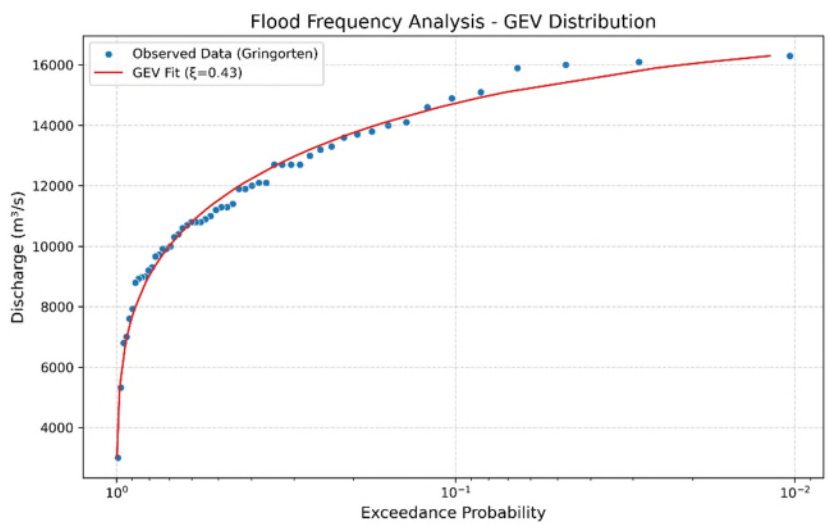


Figure 30: Generalised Extreme Value (GEV) distribution of day-mean flows during peak ice-jam staging.

The parameters f_o, f_i/f_o and μ values are drawn from previously established uniform distributions (see [IV], [V] and [VI] in Figure 29). Thousands of Monte Carlo realisations can then be generated. All computations are embedded within the Monte Carlo framework with each simulation producing a complete backwater depth profile (see [VII] in Figure 29) corresponding to a unique combination of discharge, ice-jam internal strength, friction parameters, channel slope, and channel width. Backwater results are initially calculated in nondimensional form and are subsequently converted to dimensional water-surface elevations (see [IX] in Figure 29) by adding depths to the thalweg elevations (see [VIII] in Figure 29), as elevations are required for subsequent flood mapping. Profiles of the percentiles corresponding to return periods are calculated from the ensemble of water-level elevation profiles to produce an ice-jam flood hazard curve (see [X] in Figure 29). The resulting water-surface elevations are evaluated against observed minimum and maximum ice-jam stage envelopes (see [XI] in Figure 29), established from instantaneous stage recordings (see [XII] in Figure 29), to assess whether the simulations are plausible and consistent with historical evidence.

The standalone GUI application will now be introduced. It can be launched without installing Python because all required libraries are packaged within the executable. The GUI operates within a single window (Figure 31) and allows the user to define the input directory, output directory, parameter ranges, and total number of Monte Carlo simulations. Conceptualised in Figure 32, this method generalises the point-based hazard calculations performed at the gauge by extending them to every chainage along the river reach ([IX], [X] and [XI] in Figure 29), which will be expounded upon next.

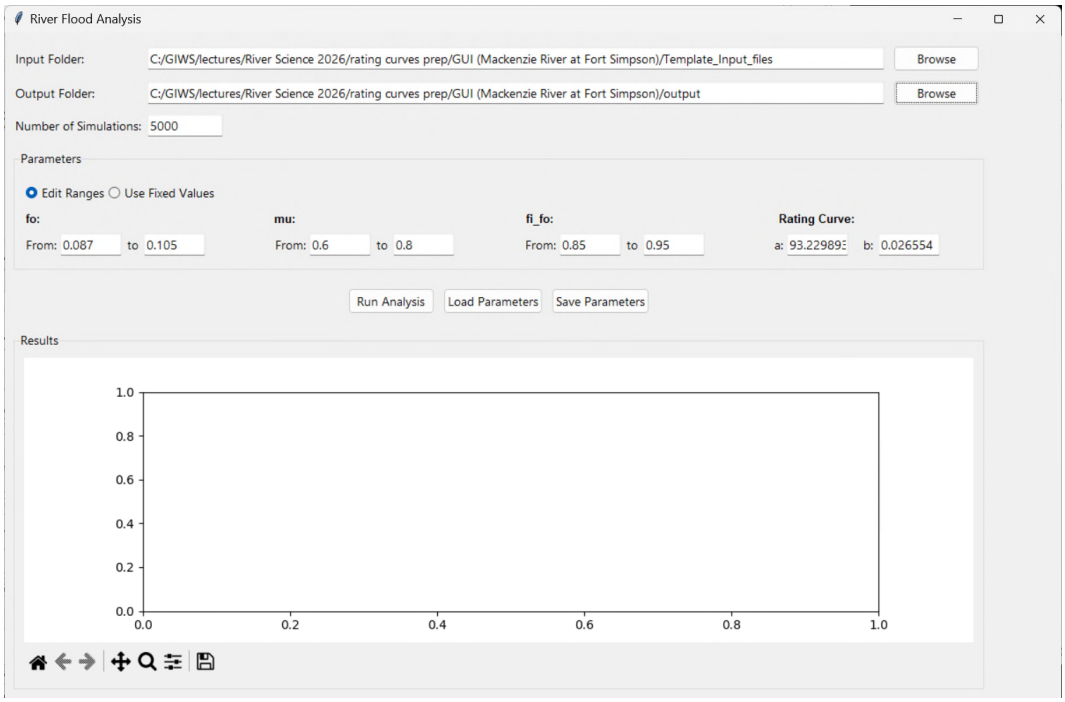


Figure 31: Graphical User Interface (GUI)

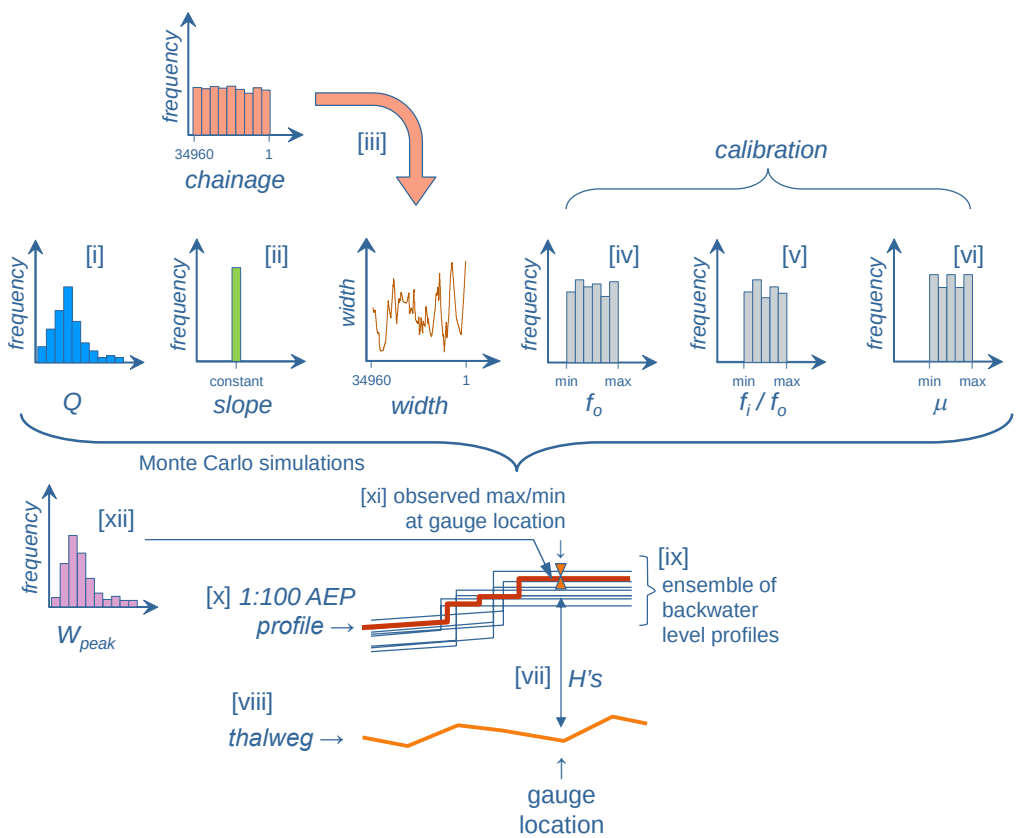


Figure 32: Conceptualisation for reach-based ice-jam flood hazard assessment.

Referring to Figure 33 (which provides more details in establishing backwater profiles for the entire reach, shown conceptually in [x] in Figure 32), for each point along the reach, the backwater level is extrapolated upstream to represent the ice-jam profile upstream of the lodgement (see [a] in Figure 33). To complete the backwater profile downstream of the ice jam (see [b] in Figure 33), the intact ice-cover rating curve (calculated in Section 3) is applied, allowing water-surface elevations to be extrapolated downstream from the gauge for a given discharge. This process produces a continuous combined profile that extends upstream into the ice-jam-affected reach and downstream into intact ice-covered conditions (see [c] in Figure 33). The entire procedure is repeated thousands of times to provide an ensemble of backwater level profiles (see [d] in Figure 33) to ensure that the resulting percentile-based hazard profiles, e.g. 1:100 AEP, are statistically stable and insensitive to random sampling variability (see [e] in Figure 33). These profiles are compared to the AEP levels stemming from the observed data recorded at the gauge (see [f] in Figure 33). The processes is repeated with adjustments to the ranges of the parameters f_o, f_i/f_o and μ until the AEP profiles coincide with the point AEP levels at the gauge, at which point the model is deemed calibrated.

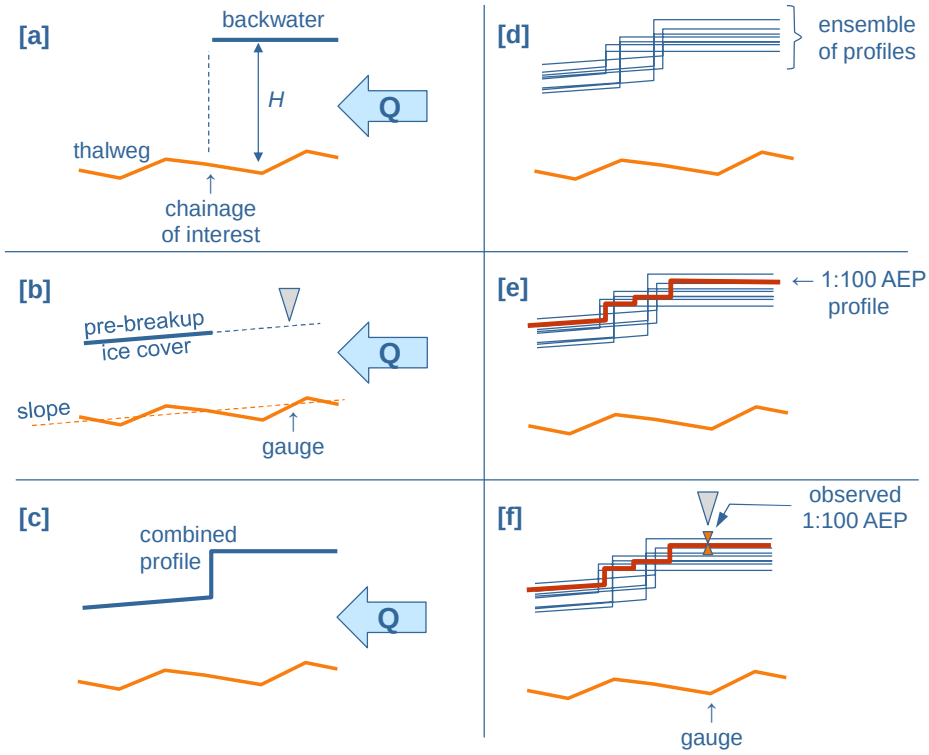


Figure 33: Method of constructing an ice-jam profile from the backwater level calculated at a point.

The primary objective of the analysis is to derive spatial hazard profiles corresponding to specific return periods, such as the 1:100 and 1:200 AEP ice-jam flood events. For each river chainage, percentile

water-surface elevations are extracted from the ensemble of simulated results, with the 99th percentile representing the 1:100 AEP profile and the 99.5th percentile representing 1:200 profile. These profiles are constructed by systematically looping through every chainage point and extracting the appropriate percentile from the distribution of simulated elevations at that location.

For the GUI, a total of six csv (comma separated variables) input files are required to execute the analysis. These include:

- discharge records used to fit the extreme-value distribution (*flow.csv*),
- gauge location information (*gauge.csv*),
- observed minimum and maximum ice-jam stages used for calibration (*gauge_levels.csv*),
- channel width and thalweg elevation data (*geomorph.csv*),
- parameter ranges defining ice-jam internal strength and friction characteristics (*parameters.csv*),
- parameters used to describe the power function of the intact-ice-cover rating curve (*rating_curve_params.csv*).

The output includes:

- *flood_levels_profile.png* – as shown in Figure 34; the highest concentration of hazard tends to occur near the Fort Simpson gauge, which is likely associated with channel narrowing and geometric constrictions that amplify backwater effects. The resulting ensembles of simulated backwater profiles provide a spatially continuous description of ice-jam flood hazard.
- *GEV_Frequency_Analysis.png* – the frequency distribution of day-mean flows during peak ice-jam staging, as shown in Figure 30
- *output.csv* – the chainage of the lodgement location, input flow and resulting backwater level elevation from each Monte-Carlo simulation.
- *percentiles.csv* – the resulting 1:100 and 1:200 AEP profiles and thalweg profile
- *river_profiles_plot.png* – shown in Figure 35, the MOCA ensemble of water-level profiles, the 1:100 and 1:200 profiles and the profiles of the thalweg and its trendline
- *thalweg_profile.png* – separate figure of the profiles of the thalweg and its trendline

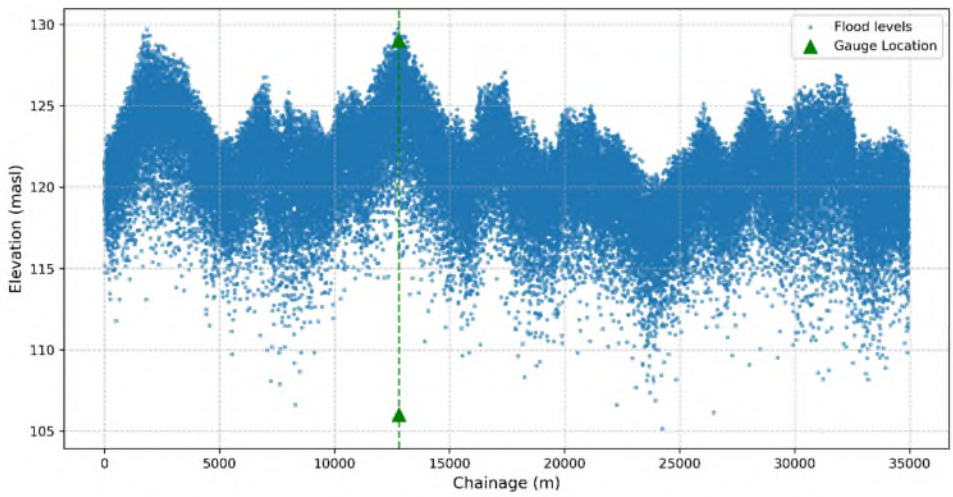


Figure 34: Point ice-jam hazard along the entire studies reach of the Mackenzie River at Fort Simpson.

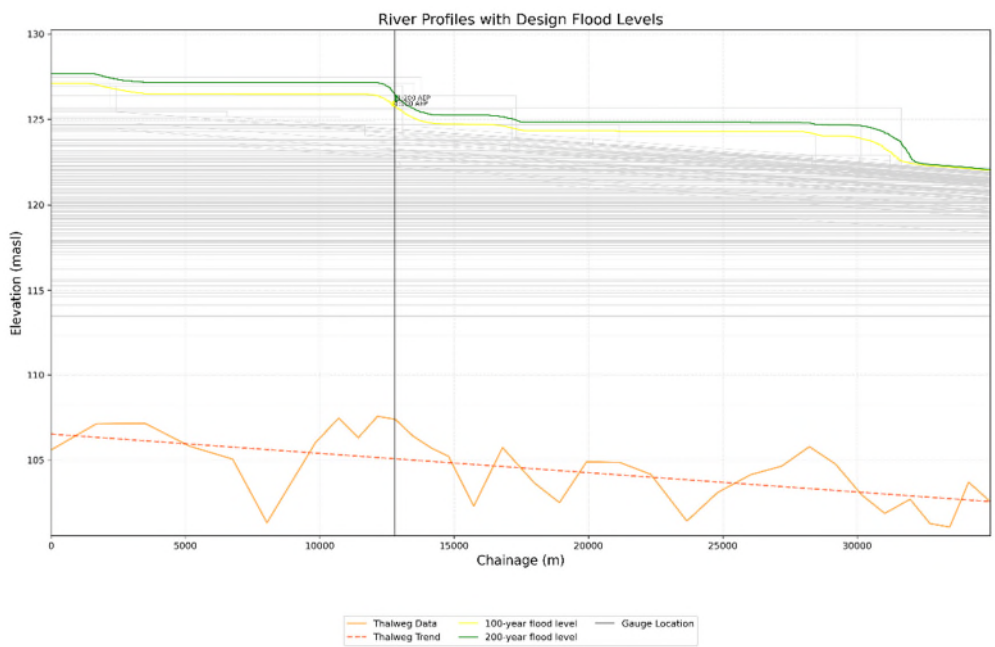


Figure 35: Ensemble of water-level profiles (gray lines) stemming from the Monte-Carlo Analysis, along with the 1:100 AEP (yellow line) and 1:200 AEP (green line) profiles and the profiles of the thalweg (solid orange line)and its trendline (dashed orange line)

Uncertainty arises from boundary effects near the downstream end of the modeled reach and from incomplete knowledge of ice-jam length and thickness. The latter's uncertainties could be addressed in future work by introducing probability distributions for ice-jam cover extent.

9. Ice-jam flood mapping of the 2021 event

This section with its accompanying video demonstrates how to create an ice-jam flood inundation map for the 2021 event by combining modeled water-surface elevations with a digital terrain model within the QGIS (https://qgis.org/) environment.

Geographical Information System step using QGIS

After launching QGIS, the first step is to create a new project so that all progress can be saved. This is done by selecting **Project → Save As** and saving the project in a designated folder, where it can be named, for example, *Fort Simpson*. Once the project is saved, the QGIS interface should be visible and ready for data import. This method is not restricted to the 2021 event but can also be followed to map scenarios yielding the 1:100 and 1:200 AEP profiles

The workflow begins in the browser panel by navigating to the "data" folder that contains the required spatial files. The digital terrain model (DTM) is first dragged and dropped into the layers panel. Once added, the DTM becomes visible on the map canvas, displaying elevation values and an associated legend. The mouse wheel can be used to zoom in and out to explore the terrain and confirm that the elevation surface has loaded correctly. Next, the transect layer is added by dragging it into the layers panel. Because the transects initially appear beneath the DTM, they should be moved above it in the layer order to ensure they are visible (see Figure 36). These transects represent cross-sections that extend across the river channel and into the adjacent floodplain and serve as the structural framework for interpolating water-surface elevations.

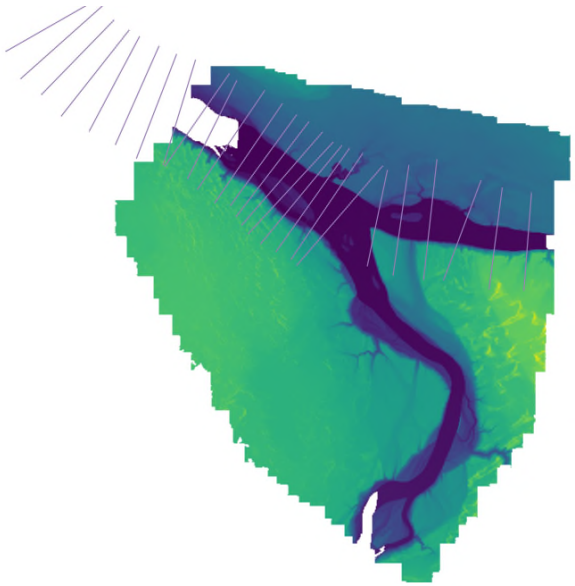


Figure 36: Transects (lines) overlaying the digital terrain model (DTM).

At this stage, the river is represented by the terrain model, and the ice-jam water-surface profile must be incorporated. The water-surface elevations are provided in the *ProfilesFortSimpson.csv* file derived from an Excel spreadsheet containing the ice-jam profile for Fort Simpson. Since each transect spans the main channel and portions of the floodplain, a specific water-surface elevation can be associated with each transect. To use these values spatially, the water-surface elevations must be joined to the transect layer. Because the *csv* file cannot be reliably added by drag-and-drop, it must be loaded using the **Open Data Source Manager**. The delimited text option is selected, the Fort Simpson profiles *csv* file is opened, and the settings are verified to ensure that field headers are present in the first line and that no geometry is assigned. Once added, the profiles table appears as a non-spatial layer in the layers panel.

Opening the attribute table of the transects layer (left panel of Figure 37) reveals fields such as transect length and identification numbers, while opening the attribute table of the profiles layer (right panel of Figure 37) shows section identifiers and water-surface elevation values. The key objective is to transfer the water-surface elevation values from the profiles table into the transects layer. This is accomplished using a table join. In the transects layer properties, a new join is created linking the profiles table to the transects layer. The common field in the profiles table is the section identifier, and the matching field in the transects table is the revised cross-section identifier. Only the field corresponding to the 2021 ice-jam event is selected for import, and a prefix such as “WSE” can be applied to clearly identify the joined water-surface elevation field. Once the join is applied, the 2021 event elevations appear in the transects attribute table, as indicated in Figure 38.

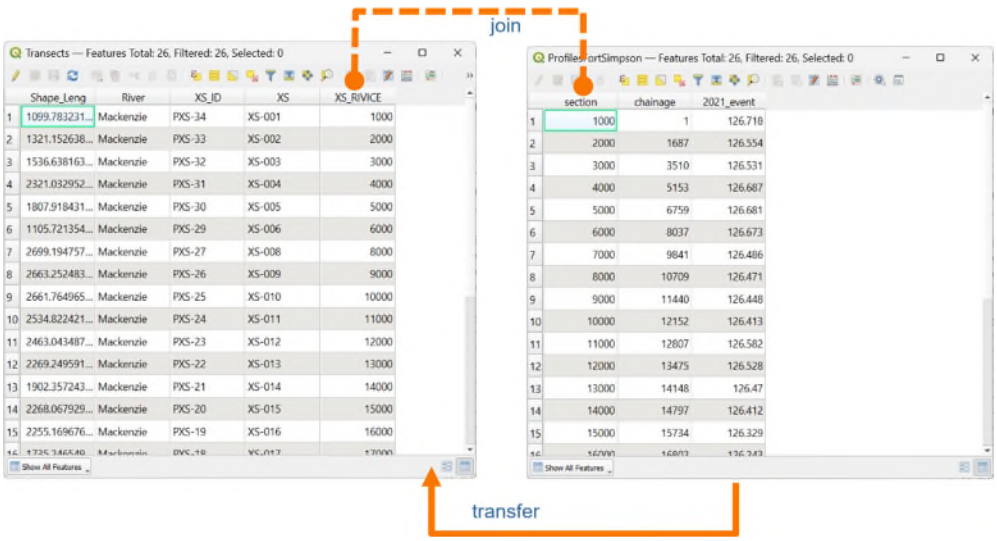


Figure 37: Joining the “section” column of the *ProfileFortSimpson* attribute table (right panel) to the “XS_RIVICE” column of the *Transects* attribute table (left column), to transfer the water-surface elevations of the 2021 event to the *Transects* table, as indicated in the next figure.

Transcects — Features Total: 26, Filtered: 26, Selected: 0

	Shape_Leng	River	XS_ID	XS	XS_RIVICE	WSE_20212021_event
1	1099.783231...	Mackenzie	PXS-34	XS-001	1000	126.718
2	1321.152638...	Mackenzie	PXS-33	XS-002	2000	126.554
3	1536.638163...	Mackenzie	PXS-32	XS-003	3000	126.531
4	2321.032952...	Mackenzie	PXS-31	XS-004	4000	126.687
5	1807.918431...	Mackenzie	PXS-30	XS-005	5000	126.681
6	1105.721354...	Mackenzie	PXS-29	XS-006	6000	126.673
7	2699.194757...	Mackenzie	PXS-27	XS-008	8000	126.471
8	2663.252483...	Mackenzie	PXS-26	XS-009	9000	126.448
9	2661.764965...	Mackenzie	PXS-25	XS-010	10000	126.413
10	2534.822421...	Mackenzie	PXS-24	XS-011	11000	126.582
11	2463.043487...	Mackenzie	PXS-23	XS-012	12000	126.528
12	2269.249591...	Mackenzie	PXS-22	XS-013	13000	126.47
13	1902.357243...	Mackenzie	PXS-21	XS-014	14000	126.412
14	2268.067929...	Mackenzie	PXS-20	XS-015	15000	126.329
15	2255.169676...	Mackenzie	PXS-19	XS-016	16000	126.243
16	1735.246549...	Mackenzie	PXS-18	XS-017	17000	126.135

Show All Features

Figure 38: Transects attribute table amended with the WSE_2021_event column.

With the transects now containing water-surface elevation information, the next step is to create a continuous water-surface raster. This is achieved by generating a triangulated irregular network (TIN) based on points sampled along each transect. Along the edges of the resulting triangles, water-surface elevations are interpolated, and the TIN is then converted into a raster surface representing water-surface elevation across the study area (see Figure 39), within the bounds of the transect extents.

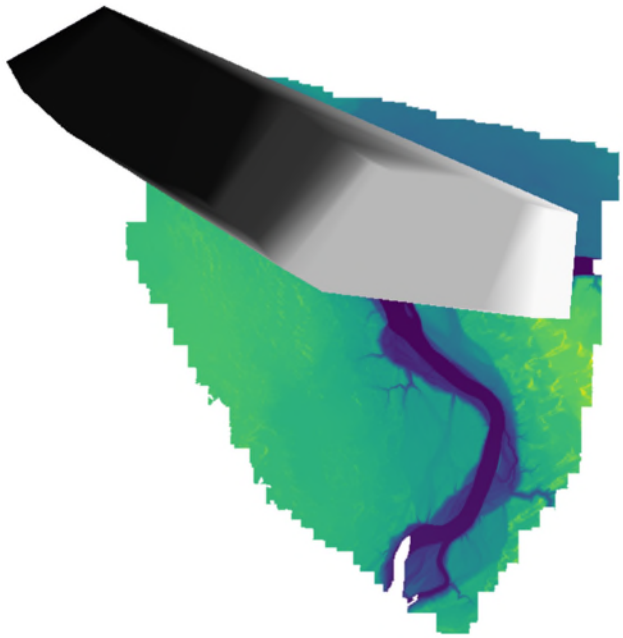


Figure 39: Water-surface elevation raster.

Using the Processing Toolbox, the **TIN Interpolation** tool is selected. The transects layer is used as the interpolation input, with the water-surface elevation field specified as the attribute to interpolate. The transects are treated as structural lines to guide the interpolation. To restrict the interpolation to the area covered by the transects, the output extent is calculated directly from the transects layer. A pixel size of 10 metres is selected to balance computation speed and spatial resolution. Running the tool produces a raster representing the interpolated ice-jam water-surface elevation.

Flood extent and depth are determined by comparing the interpolated water-surface raster with the digital terrain model. Using the Raster Calculator, the DTM is subtracted from the water-surface raster to create a temporary difference raster. Where this difference is positive, the water surface lies above the terrain and flooding occurs; where it is zero or negative, the ground is not flooded (indicated in Figure 40 and explained conceptually below in subsection "*Conceptualisation of the flood depth mapping*").

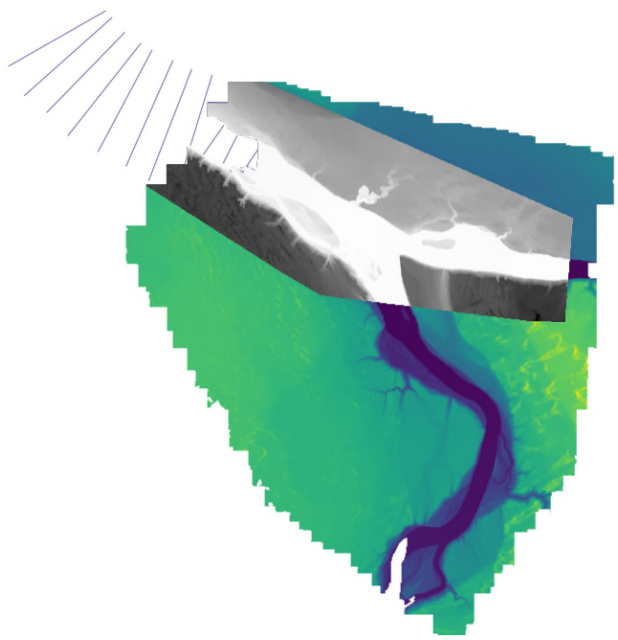


Figure 40: Raster of differences between the water-surface elevation (WSE) raster and the digital-terrain model (DTM) raster.

A flood mask is then created by applying a conditional operation to the difference raster. Cells with values greater than zero are assigned a value of one, indicating flooded areas, while cells with values less than or equal to zero are assigned a value of zero, indicating non-flooded areas. This binary raster clearly delineates the flood extent (see Figure 41 and conceptual explanation in subsection "*Conceptualisation of the flood depth mapping*" below).

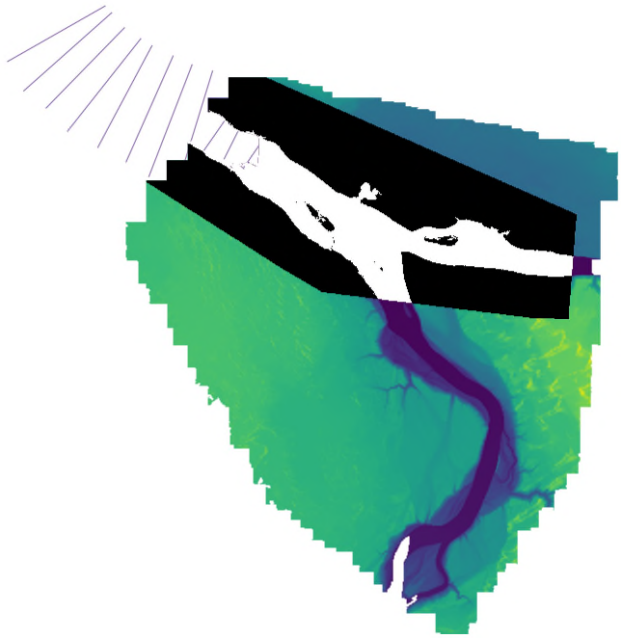


Figure 41: Mask of flooded extent.

Finally, flood depths are calculated by multiplying the difference raster by the flood mask (see Figure 42 and conceptualisation in the next subsection). This operation retains positive depth values in flooded areas while forcing non-flooded cells to zero. The resulting depth raster is displayed by turning off other layers and applying an appropriate symbology. A blue color ramp is used to represent flood depth, and zero-value cells are made transparent.

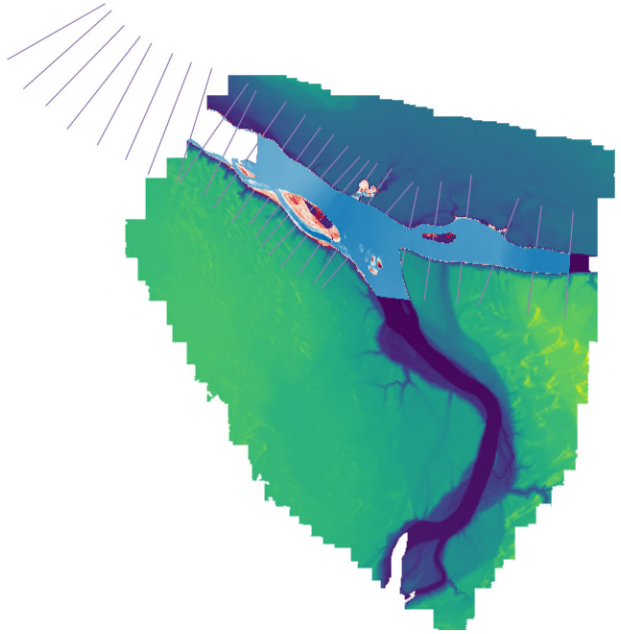


Figure 42: Flood depth raster.

The final map illustrates flood depths associated with the 2021 ice-jam event (Figure 43). Fort Simpson, which is situated on an island, is clearly visible within the inundation pattern. The deepest water occurs within the main river channel, while surrounding floodplain areas also experience inundation, highlighting the spatial extent and severity of ice-jam flooding during the event.

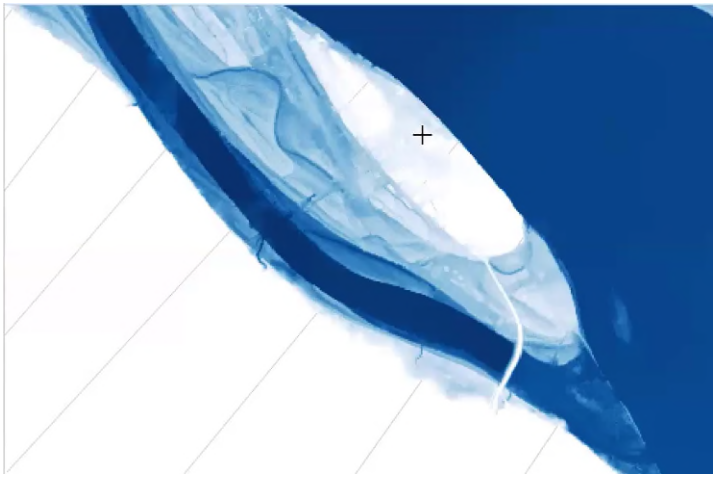


Figure 43: Map of flood depths at Fort Simpson from the 2021 ice-jam flood event.

Conceptualisation of the flood depth mapping

Using the concepts illustrated in the figures from Figure 44 to Figure 48, the development of the water-surface elevation (WSE) raster can be described as a continuous geospatial process that moves from a conceptual understanding of river hydraulics to a spatially explicit flood-depth map. The process begins with a side view of the river reach (see Figure 44), where the thalweg is shown with an overall longitudinal slope that controls downstream flow conditions. Superimposed on this longitudinal profile is the ice-jam water-surface elevation profile, which represents how water levels rise above the normal river surface due to ice obstruction and backwater effects. This side-view perspective establishes the fundamental relationship between channel slope, ice-jam conditions, and resulting water levels that must be projected into plan view for flood mapping (explained next).

Referring to Figure 45, the workflow then shifts to a plan view of the river system, where the spatial geometry of the river channel is shown from above, together with the lateral extent of the floodplain that may become inundated during the ice-jam event. Within this plan view, the previously defined WSE profile is conceptually extended along the middle of the channel following the centreline, ensuring that the longitudinal variation in water-surface elevation is spatially aligned with the river's primary flow path. Discrete water-surface elevation points are then defined along this centreline at locations corresponding to

locations where the cross-sectional transects intersect the channel, with the transects drawn perpendicular to the centreline so that they properly capture lateral variations across the channel and into the floodplain.

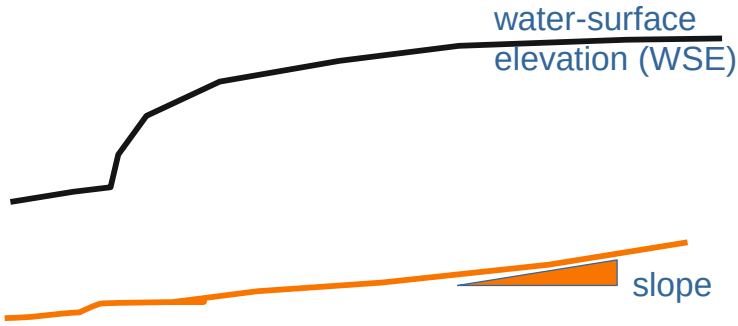


Figure 44: Side view of river reach indicating the thalweg slope and the ice-jam water-surface elevation profile.

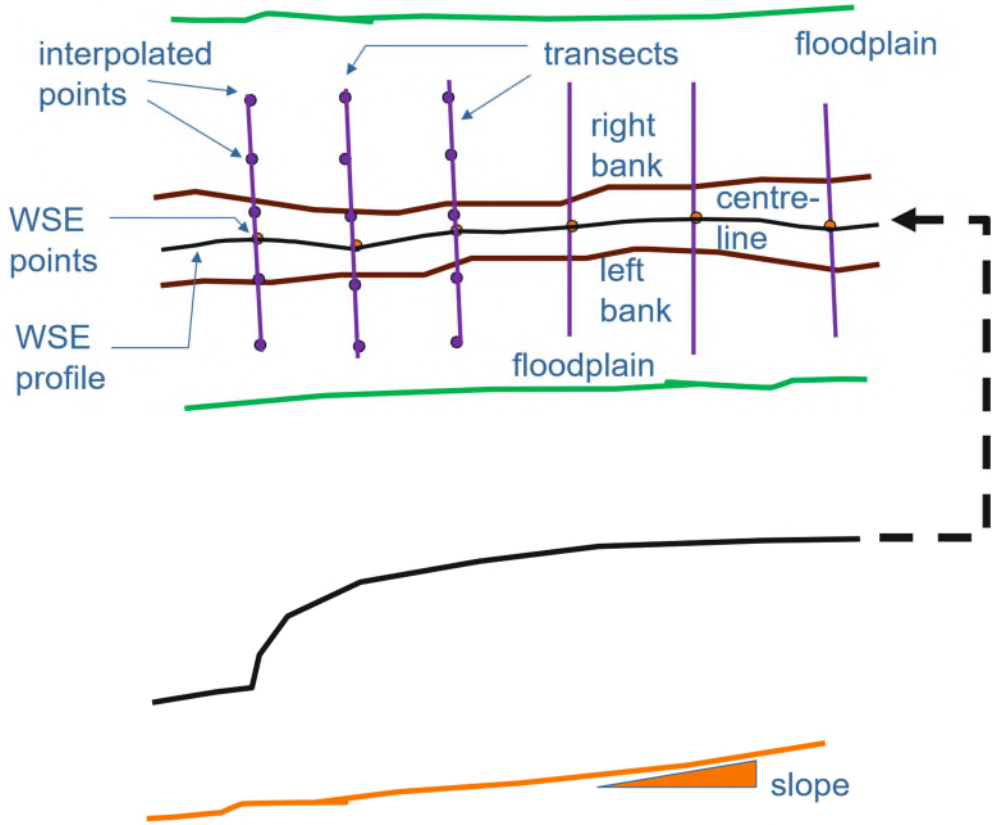


Figure 45: Plan view (top) and side view (bottom) of river reach.

Once the centreline elevations have been established, the WSE values are interpolated horizontally along each transect from the channel outward, creating a set of interpolated elevation points that represent the

water surface across the full width of the floodplain at each cross section. These interpolated points are then connected using a triangulated irregular network (TIN) (see Figure 46), in which adjacent points are linked to form a mesh of non-overlapping triangles that collectively describe a continuous water-level elevation surface over the study area.

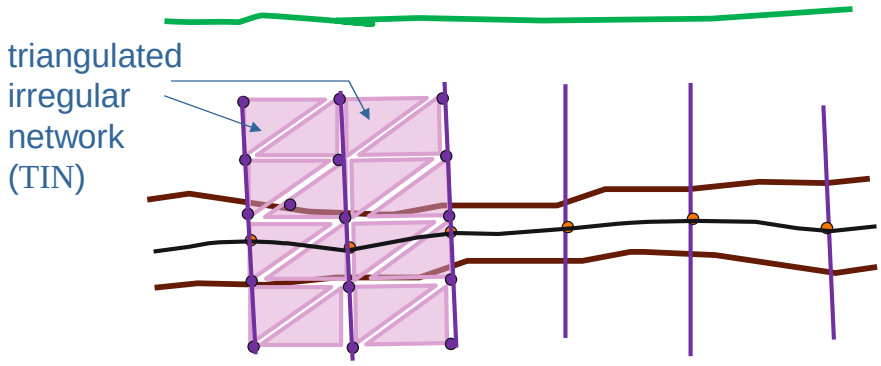


Figure 46: Development of a triangulated irregular network (TIN).

To ensure that this surface can be converted into a raster format suitable for spatial analysis, additional points are further interpolated along the edges of the triangles so that there is sufficient point density to populate an orthogonal grid (Figure 47).

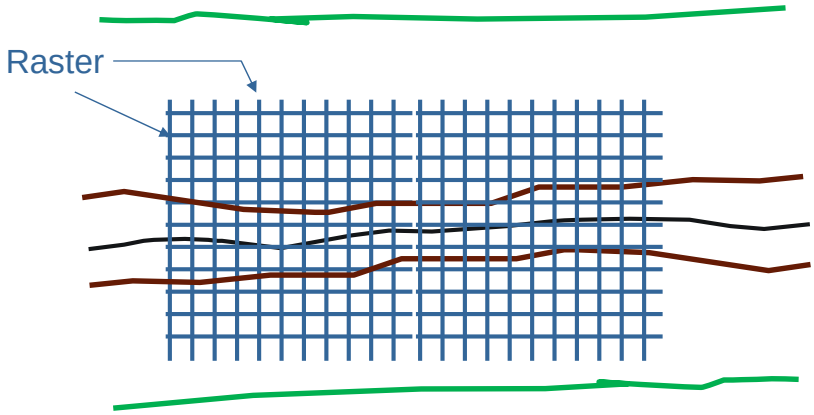


Figure 47: Developing the water-surface elevation raster.

The resulting WSE raster can then be examined in a cross-sectional view looking downstream, where it is shown intersecting with the digital terrain model (DTM) that represents the ground surface (top panel of Figure 48). Subtracting the DTM from the WSE raster yields a temporary difference raster, commonly referred to as temp_diff, in which positive values indicate that the modeled water surface lies above the ground and negative values indicate that the ground surface remains above the water surface (middle panel of Figure 48). From this difference raster, a binary mask named “flooded” is calculated such that

cells with temp_diff greater than zero are assigned a value of one, indicating flooded areas, while cells with temp_diff less than or equal to zero are assigned a value of zero, indicating areas that remain dry. Flood depths are then calculated only where flooding occurs, by multiplying the temp_diff raster by the flooded mask so that non-flooded areas are forced to zero depth (bottom panel of Figure 48). The resulting flood-depth raster provides a spatial representation of inundation depths across the channel and floodplain, which can then be mapped and analyzed to support interpretation of the ice-jam flood event.

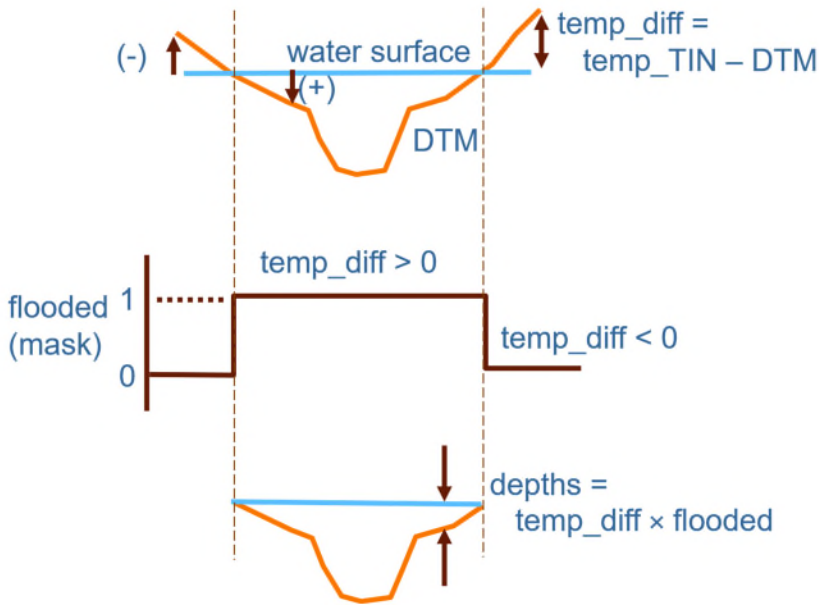


Figure 48: Cross-sectional view of river channel indicating conceptually the GIS steps to derive flood depths.

10. Extrapolation of ice-jam flood hazard (to Jean Marie River)

This section, with the accompanying video, demonstrates how ice-jam flood hazard estimates developed for Fort Simpson are extrapolated to an ungauged location at Jean Marie River. The analysis builds directly on Section 7 where the ice-jam staging calibration for Fort Simpson is completed. Here the focus is on explaining the conceptual basis for transferring ice-jam flood hazard estimates between sites along the Mackenzie River, from a gauged site (Fort Simpson) to an ungauged site (Jean Marie River).

In the earlier Fort Simpson analysis, a flow frequency distribution corresponding to discharges at maximum ice-jam backwater stages was derived (Section 4). At the gauge site, channel slope and channel width were determined, and three key model parameters were calibrated: f_o = composite friction factor representing the combined resistance of the riverbed and ice, f_i/f_o = ratio of ice friction to the composite friction, μ = internal ice-jam strength parameter for the 2021 ice-jam flood event (Section 5). These

parameters were then randomly sampled within a Monte Carlo framework, producing an ensemble of backwater depths (Section 7). The backwater depths were then added to a base elevation to obtain water-surface elevations, which could be used directly for flood hazard mapping.

Observed return periods for ice-jam backwater levels at Fort Simpson were available from the data, and three statistical distributions were fitted to the observations. The minimum and maximum envelopes from these best-fitting distributions were used to constrain the Monte Carlo simulations by adjusting parameter ranges until the modeled hazard curve fell approximately within the observed bounds. This calibration process was carried out at the gauged site, but the same general methodology can be applied at an ungauged location, provided that suitable assumptions are made.

Referring to Figure 49, for the extrapolation to Jean Marie River, the calibrated parameterization and the same parameter ranges derived at Fort Simpson are retained ([4], [5] and [6] in Figure 49). Flow inputs, however, are obtained from Strong Point, which is another gauged location on the Mackenzie River. Fort Simpson is located downstream, while Jean Marie River lies approximately 50 kilometres upstream. Strong Point provides discharge and water-level data during the ice-covered period, and a flow-frequency analysis was previously performed for flows associated with peak ice-jam staging ([1] in Figure 49). Random discharges drawn from this distribution are used as inputs to the Monte Carlo simulations.

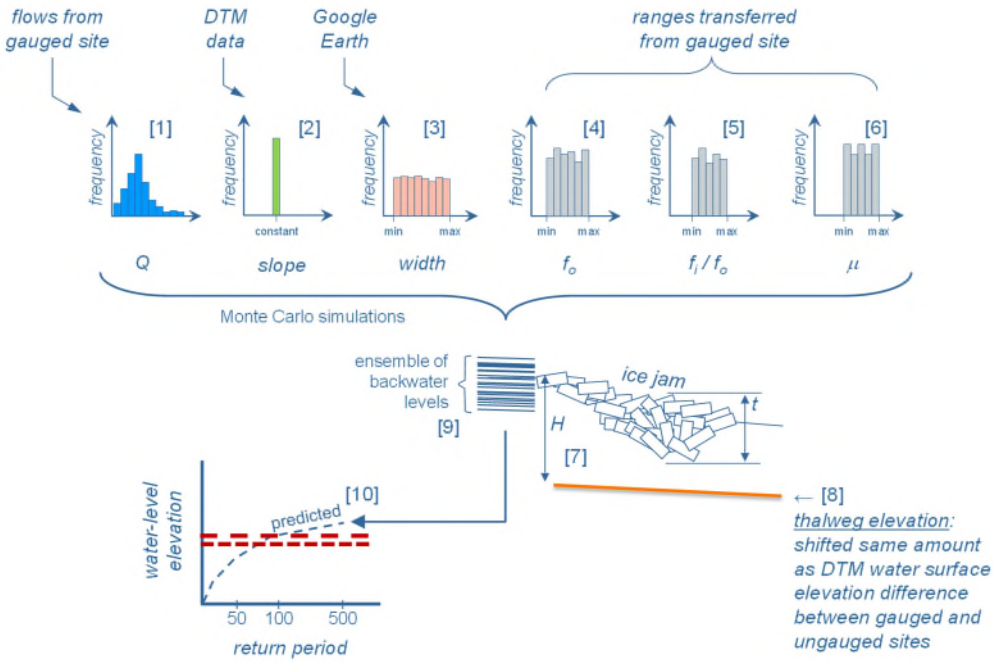


Figure 49: Conceptualisation for extrapolating ice-jam flood hazard assessment from the Fort Simpson gauge to Jean Marie River.

The large source of uncertainty in the extrapolation arises from estimation of the channel slope ([2] in Figure 49) at Jean Marie River. Because detailed bathymetric data are unavailable, the slope must be derived from the digital terrain model rather than from surveyed channel geometry (see Figure 50). Channel slope is estimated using LiDAR data by extracting elevations along the water surface of the Mackenzie River within the study reach. From these data, an average slope of approximately 0.0000686 is calculated and adopted for the model simulations.

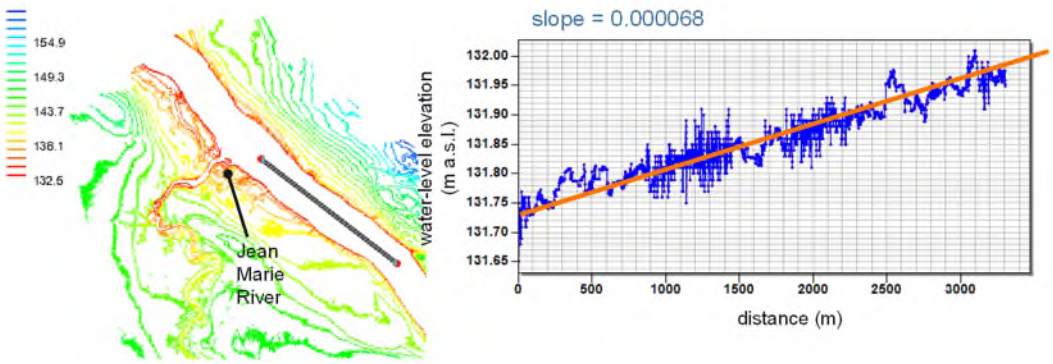


Figure 50: Determining slope from the digital terrain model at jean Marie River.

Unlike the Fort Simpson reach-scale assessment, this extrapolation is carried out at a single point rather than spatially along a reach. Moving from Excel file *(2) ice-jam_staging_calibration (FortSimpson).xlsx* to *(3) ice-jam_staging_extrapolate (JeanMarieRiver).xlsx*, as stepped through in the video, the channel width near the community of Jean Marie River is estimated from Google Earth to be approximately 900 m, with the narrowest section measuring roughly 700 m. To represent this uncertainty, channel width is treated as a random variable and sampled uniformly between 700 and 900 metres in the Monte Carlo framework (see [3] in Figure 49).

The thalweg elevation ([viii] in Figure 49) is the largest source of uncertainty and requires additional assumptions. At Fort Simpson, the water-surface elevation from the DTM is approximately 118.6 metres, while at Jean Marie River it is approximately 131.7 metres, representing an increase of about 13.1 metres. To estimate the thalweg elevation at Jean Marie River, the Fort Simpson gauge elevation is increased by 13.1 metres, assuming that the elevation difference between the two locations is consistent along the reach. This adjusted thalweg elevation is applied uniformly across all Monte Carlo realisations and added to the backwater depths ([7] in Figure 49) to yield the ensembles of backwater level elevation ([9] in Figure 49).

Some high-water marks are available near Jean Marie River that provide useful constraints. The ice-jam flood hazard results suggest that the observed high-water marks correspond approximately to return periods between the 1:100 and 1:200 AEP levels (provided in Figure 51 and shown conceptually in [10] of Figure 49). Increasing the number of Monte Carlo realizations to approximately 10,000 substantially smooths the hazard curve and yields more stable percentile estimates.

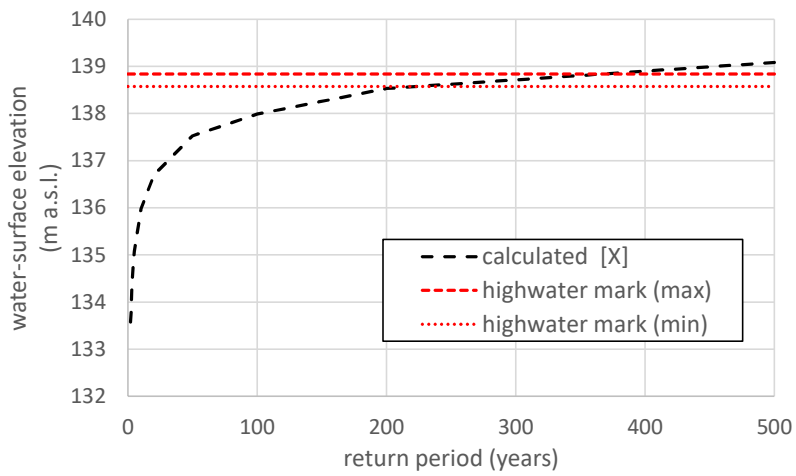


Figure 51: ice-jam flood hazard curve for Jean Marie River.

Uncertainty in thalweg elevation remains a dominant limitation of the analysis. A key assumption in this extrapolation is that the increase in thalweg elevation from Fort Simpson to Jean Marie River is equal to the increase in water-surface elevation during ice-jam conditions. This assumption may not be strictly valid. Simple backwater calculations using Manning's equation, assuming similar bed roughness conditions, indicate that flow depths at Jean Marie River may be proportionally greater than those at Fort Simpson. As a result, the extrapolated hazard estimates may be slightly underestimated.

For flood mapping applications at Jean Marie River, the extrapolated hazard curve can be combined with the estimated channel slope. A simplified trapezoidal cross-section can be assumed, and water-surface elevations corresponding to selected return periods, such as the 1:100 and 1:200 AEP levels, which can be projected upstream and downstream along the slope. These elevations can then be intersected with floodplain topography using transects to generate flood inundation and depth maps.

Epilogue

This user manual has demonstrated that empirically formulated, reach-based and probabilistic approaches provide a practical pathway for assessing ice-jam flood hazards in data-sparse northern river systems,

while explicitly acknowledging uncertainty and the stochastic nature of river ice processes. By combining simplified physical relationships, Monte Carlo simulation, and fluvial geomorphological information, the methodology progresses from point-based analysis at a hydrometric gauge to spatially continuous hazard profiles along an entire river reach, thereby culminating in a reach-based extrapolation of ice-jam flood hazard from a gauged location to an ungauged reach. This extrapolation represents a central outcome of the framework and enables flood hazard assessment at locations where conventional hydraulic modelling is challenged due to the limitation or absence of data.

While the framework is intentionally streamlined to maximize transferability and transparency, its application has highlighted several physical processes and river characteristics that warrant further development to extend the robustness and realism of future assessments. One important avenue for future work is the explicit incorporation of abstractions of water from the main channel to represent overbank flow into adjacent floodplains in river reaches characterized by very low banks and wide floodplains. Such conditions are common along large, low-gradient rivers, including the Athabasca River at Embarras and Old Fort, or the Peel River at Fort McPherson, where floodwaters can readily spill out of the main channel during ice-jam events and be temporarily stored across extensive floodplain areas. In these settings, neglecting lateral losses from the main channel can lead to systematic overestimation of backwater levels, whereas a parameterized abstraction of discharge offers a practical means of mimicking floodplain storage within an otherwise one-dimensional empirical framework.

Abstractions should also be considered for river sections with a high off-channel storage capacity arising from secondary channels, off-channel lakes, and low-gradient tributaries. These geomorphic features act collectively as buffers during ice-jam flooding by storing backwater away from the main channel, thereby attenuating peak stages, and their net hydraulic effect can be reasonably simulated through discharge abstractions that are triggered once overbank conditions are exceeded. Such representations are particularly relevant for anabranching rivers and deltaic or pre-deltaic reaches where flow partitioning becomes increasingly important under high stages.

Additional research is also recommended to explicitly account for longitudinal variations in riverbed slope along a single river system. Many large rivers, such as the Athabasca River, exhibit transitions from relatively steep riverbeds in their upper reaches to progressively flatter slopes downstream, as observed at Fort McMurray, and these changes can significantly influence ice-jam formation and backwater staging severity. Incorporating spatially variable slopes into the empirical and Monte Carlo framework would improve representation of backwater propagation across such transitions. Further methodological

refinement is also needed for rivers whose riverbed gradients change from positive to negative in the downstream flow direction, as occurs along the Mackenzie River at Tsiigehtchic.

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Disclaimer

Please note, this user manual and its associated videos, data sets and coding scripts, are the product of an applied science endeavour. The methodologies used in generating the research results contained herein are novel, and further testing is required to verify the methodology's robustness. While every effort has been made to ensure the accuracy and completeness of the information contained herein, neither the authors nor the University of Saskatchewan make any guarantee, nor do they assume any liability for any errors or omissions. The information provided is on an 'as is' basis and the authors and University of Saskatchewan disclaim all warranties, express or implied, including but not limited to, implied warranties of merchantability or fitness for a particular purpose.

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